

FLEXIBLE IMPUTATION OF INCOMPLETE NETWORK DATA ^{*}

Ge SUN[†] Weisheng ZHANG [‡]

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Abstract

Sampled network data are common in empirical research because collecting full network information is costly, but using sampled networks can lead to biased estimates. We propose a flexible imputation method for sampled networks and show that downstream empirical analysis based on imputed networks yields consistent parameter estimates. We propose a nonparametric method that imputes missing network links by combining a projection onto covariates with a local two-way fixed-effects regression. The imputation method avoids parametric assumptions, does not rely on low-rank restrictions, and flexibly accommodates both observed covariates and unobserved heterogeneity. We establish entrywise convergence rates for the imputed matrix and prove the consistency of GMM estimators based on the imputed network. We further derive the convergence rate of the corresponding estimator in the linear-in-means peer-effects model. Simulations show strong performance of our method both in terms of imputation accuracy and in downstream empirical analysis. We illustrate our method with an application to the microfinance network data of [Banerjee et al. \(2013\)](#).

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[†]University of Notre Dame: gsun4@nd.edu.

[‡]University College London: weisheng.zhang.21@ucl.ac.uk.

1 Introduction

There is a large and growing literature studying how social networks shape individuals’ behavior and economic outcomes. Examples include how social interactions affect information diffusion (Banerjee et al., 2013; Beaman and Dillon, 2018), test scores (Sacerdote, 2001), and demand for financial assets (Conley and Udry, 2010; Cai, Janvry and Sadoulet, 2015). Despite extensive empirical work, many conclusions are undermined by the fact that researchers rarely observe the full network. Since collecting full network data is costly, applied work typically relies on sampled networks instead. However, Chandrasekhar and Lewis (2011) demonstrate that using sampled networks in place of the full network may lead to biased estimates of regression coefficients or GMM parameters.

To overcome this limitation, we propose a flexible imputation method for sampled networks and show that downstream empirical analysis using the imputed network delivers consistent parameter estimates. Our proposed method is designed for networks generated by dyadic formation processes and observed through egocentric sampling, one of the most commonly used designs in which researchers randomly sample a set of nodes and ask all their friends, not only within the sample but in the entire network. The proposed method separately imputes the part of each missing link that is explained by covariates and the remaining variation not captured by them: the former via projection onto observed covariates, and the latter using local two-way fixed-effects regression. The method has several advantages over existing approaches, making it particularly well suited to the social networks studied in economics.

First, the proposed method avoids parametric assumptions. Many imputation approaches, such as Chandrasekhar and Lewis (2011), Breza et al. (2020), and Boucher and Houndetoungan (2020), rely on knowing the functional form of the network formation model. In practice, however, applied researchers rarely have such prior information. By contrast, our approach is fully nonparametric and therefore robust to misspecification in network formation models.

Second, our imputation method does not rely on the low-rank assumption. A common approach in the statistical network literature is to assume a low-rank network formation model and exploit this structure for imputation (Li et al., 2023; Cai, Cai and Zhang, 2016). However, this assumption fails when the network formation process is nonlinear. For example, the widely used network formation model in Graham (2017) does not, in general, have a low-rank structure. Instead, building on ideas from Freeman and Weidner (2023) and Feng (2023), we approximate the nonlinear network formation function using local two-way fixed-effects regression, without imposing global low-rank assumptions, making our method applicable to a wide class of nonlinear network formation processes.

Third, our method leverages observed covariates and accommodates unobserved heterogeneity in link formation. Social networks typically exhibit homophily, where individuals with similar characteristics are more likely to form links, as well as degree heterogeneity, where some individuals tend to have more connections than others. Therefore, an effective imputation procedure should take advantage of available covariate information while also accommodating unobserved heterogeneity in link formation. Our approach achieves both objectives without imposing assumptions on the joint distribution of latent factors and observed covariates.

Our general network formation setting leads to a technical challenge: the latent factors cannot be identified when the network formation model is unknown and potentially nonlinear. As a result, we cannot first estimate the latent factors, as is commonly done in linear factor models, and then estimate the formation process to impute the missing links. To address this difficulty, we build on the idea, first developed by Zhang, Levina and Zhu (2017), that similarity in observed connections can reveal similarity in latent factors. Using the observed subnetwork, we first construct a pseudo-distance that serves as a proxy for latent distance. Then, for each unsampled individual, we find sampled individuals with small pseudo-distances and use their observed connections to impute that unsampled individual’s missing links.

The main theoretical result for imputation is Theorem 2, which, as the network size and the number of sampled nodes go to infinity, establishes the sup-norm convergence rate of the proposed imputation and provides uniform asymptotic orders for the bias and variance of the imputation error. Such sup-norm control over both the bias and variance is new to the literature and is particularly useful when analyses require accurate predictions of the conditional probability of forming a specific link. Because our method relies on a pseudo-distance that approximates latent similarity, the imputation inevitably introduces approximation error. Theorem 2 shows that this error has an asymmetric impact on the imputation error: the leading asymptotic order of the bias does not depend on the pseudo-distance error, whereas the leading order of the variance may be affected by it.

We further show in Theorem 3 that, for downstream empirical analyses, replacing the sampled network with our imputed network yields consistent estimates of the GMM parameters. This result is particularly relevant in empirical applications, where interest typically focuses on the consistency of estimators constructed from the imputed network rather than on the accuracy of the imputed links themselves. In addition, since our imputation is robust to misspecification of the network formation process and does not require low-rank assumptions, the resulting GMM estimator is also more robust than existing approaches.

Characterizing how imputation error affects the estimation of model parameters is technically challenging. The difficulty arises not only because the nonparametric imputation error for each link is nonclassical, but also because empirical analyses typically aggregate links into network statistics, which leads to aggregated measurement error. To examine how imputation errors influence parameter estimation, we study the linear-in-means peer-effects model and derive the convergence rate of the estimator based on the imputed network in Theorem 4. In addition, the theorem implies that the bandwidth optimal for link-level imputation is not optimal for the downstream GMM estimator: the two have different asymptotic orders. In particular, the bias of the link-level imputation error plays a more prominent role than variance in determining the estimator’s convergence rate.

We provide extensive simulation evidence to evaluate the finite-sample performance of our imputation method. First, we compare its imputation accuracy with alternative approaches and show that our method outperforms them across a wide range of sampling rates and network sparsity levels. We then evaluate the performance of our method in empirical analysis, including regressions based on network statistics and the linear-in-means peer-effects model. These simulations demonstrate that the estimates of regression coefficients and GMM parameters based on our imputed networks exhibit strong finite-sample performance. Finally, we illustrate the practical usefulness of the proposed method through an empirical application to the study of how social networks affect microfinance adoption (Banerjee et al., 2013).

Related literature We contribute to the literature on estimating model parameters with sampled network data by developing an imputation method that allows for a fully nonparametric network formation model. Chandrasekhar and Lewis (2011) show that models estimated using sampled networks are generally biased. They propose an analytical correction for certain special cases and, for more general settings, they estimate a parametric network formation model and conduct GMM using simulated moments. Boucher and Houndetoungan (2020) study the estimation of linear-in-means peer-effects model when the network data are corrupted, taking egocentrically sampled networks as a special case. Their approach relies on obtaining a consistent estimator of the distribution of the network, which is typically available only under parametric specifications. Herstad (2023) also examine peer-effects estimation with missing links, focusing on settings with large networks and relying on the network formation model of Graham (2017). A key distinction relative this line of work is that our method does not rely on parametric specifications of the network formation process and imposes no restrictions on the form of unobserved

heterogeneity. Therefore, parameter estimation based on our imputed network are robust to misspecification. In addition, our approach for estimating GMM parameters differs from that of [Chandrasekhar and Lewis \(2011\)](#), who construct moments by integrating over the missing network data. Instead, we directly plug the imputed network into the moment conditions to avoid heavy computational burden.

Another related line of research studies the recovery of missing links under partial network data or even in the absence of network data. [Breza et al. \(2020\)](#) demonstrate that aggregated relational data can help recover network structure without observing any links. [de Paula, Rasul and Souza \(2025\)](#) study how network ties can be identified from panel data. [Thirkettle \(2019\)](#) show that network statistics are set identified under partially sampled networks, although their analysis relies on a strategic network formation model, whereas ours is based on a dyadic formation framework. Other sampling designs have also been studied—for example, induced subgraphs in [Chatterjee \(2015\)](#) and censored networks in [Griffith \(2022\)](#).

Our work also relates to the growing literature on estimation with mismeasured network data, as empirical analysis based on imputed networks inevitably inherits the errors introduced in the imputation stage. For peer-effects models, [Lewbel, Qu and Tang \(2024\)](#) show that when the number of misclassified links grows at a rate slower than the network size, the 2SLS estimator of [Bramoullé, Djebbari and Fortin \(2009\)](#) remains consistent and standard inference methods remain valid. [Lewbel, Qu and Tang \(2025\)](#) propose an adjusted 2SLS estimator for peer-effects models under i.i.d. link misclassification. [Hardy et al. \(2024\)](#) also study link misclassification under i.i.d. errors and obtain consistent treatment effects using an expectation-maximization algorithm. [Cai \(2022\)](#) examine linear regression on centrality measures under link-level i.i.d. Bernoulli errors, and propose bias-correction and inference methods for OLS estimators. The setting we consider is technically more challenging than those in the existing literature. Our imputed networks contain errors that are dependent across links and may have bias of the same order as their standard deviation. Our paper contributes to the literature by characterizing the convergence rate of linear-in-means peer-effects estimators in settings that allow for far more general forms of network measurement error.

Methodologically, our approach is related to causal inference using panel data with staggered adoption, as the two settings share similar missing-data structures and face similar challenges associated with unobserved heterogeneity. This line of work includes the rapidly growing literature on synthetic control and causal matrix completion, such as [Arkhangelsky et al. \(2021\)](#), [Athey et al. \(2021\)](#), [Bai and Ng \(2021\)](#), and [Agarwal et al. \(2023\)](#), among others. Most of these methods assume that the potential outcomes follow a low-rank structure plus idiosyncratic noise. The studies most directly connected to ours are in the recent literature on nonlinear factor models in panel-data causal inference, particularly the methods of [Feng \(2020\)](#), [Feng \(2023\)](#), and [Deaner, Hsiang and Zelenev \(2025\)](#), which also allow for nonparametric and nonlinear factor structures. Our local two-way fixed-effects regression is analogous to a local linear regression and therefore achieves a second-order approximation error. In contrast, the estimator in [Deaner, Hsiang and Zelenev \(2025\)](#) corresponds to a local constant regression and attains only a first-order approximation error. The local PCA methods of [Feng \(2020\)](#) and [Feng \(2023\)](#) can be viewed as local polynomial regressions that fully exploit smoothness of nonlinear functions. Our approach differs from local PCA in that it does not require specifying the latent factor dimension and is more robust to the misspecification of smoothness assumptions. More broadly, unlike most methods in this strand of the literature, our approach flexibly incorporates observed covariates when imputing missing values.

Finally, our analysis is connected to several strands of the econometrics and statistics literature. The use of pseudo-distance as a measure of latent similarity was first developed by [Zhang, Levina and Zhu \(2017\)](#) and has since been applied in a range of contexts, including nonparametric graphon estimation ([Zhang, Levina and Zhu, 2017](#); [Zelenev, 2020](#)), controlling for unobservables using network

data (Auerbach, 2022), and nonlinear factor models (Feng, 2020, 2023; Beyhum and Mugnier, 2024; Mugnier, 2025; Deaner, Hsiang and Zelenev, 2025). Our analysis is also related to the dyadic network formation literature (Graham, 2017; Dzemski, 2019; Chen, Fernández-Val and Weidner, 2021; Ma, Su and Zhang, 2022) and the graphon estimation literature (Gao, Lu and Zhou, 2015; Xu, 2018). Kitamura and Laage (2024) consider covariate-assisted graphon estimation similar to ours, but they focus on stochastic block models, whereas we focus on more general nonlinear link-formation process. Finally, extracting the component of the network explained by covariates relies on the dyadic nonparametric regression developed by Graham, Niu and Powell (2021).

The rest of the paper is organized as follows. Section 2 introduces the framework and estimation procedure. Section 3 provides statistical guarantees for the imputation. Section 4 presents theoretical properties of the GMM estimators based on the imputed network. Section 5 presents Monte Carlo results for both the imputation and the downstream empirical analysis. The empirical application to Banerjee et al. (2013) is given in Section 6. Section 7 concludes.

2 Method

2.1 Network Formation Model

Let $A \in \{0, 1\}^{N \times N}$ denote the adjacency matrix of an unweighted network, where $A_{ij} = 1$ if there is a link between i and j and $A_{ij} = 0$ otherwise. We impose $A_{ij} = A_{ji}$ for all i, j and $A_{ii} = 0$, so that the network is symmetric and has no self-loops. For each individual i , let $X_i \in \mathbb{R}^{d_x}$ denote the observed covariates (e.g., gender, education, and other demographic characteristics), and let $\xi_i \in \mathbb{R}^{d_\xi}$ denote the latent factors. We write $X = \{X_i \mid i = 1, 2, \dots, N\}$ and $\xi = \{\xi_i \mid i = 1, 2, \dots, N\}$ for brevity. The network A is generated according to the following process. Conditional on (X, ξ) , the links $\{A_{ij}\}_{1 \leq i < j \leq N}$ are generated independently with probability $\mathbb{P}(A_{ij} = 1 \mid X, \xi) = f(X_i, \xi_i, X_j, \xi_j)$, where $f : \mathbb{R}^{d_x + d_\xi} \times \mathbb{R}^{d_x + d_\xi} \rightarrow [0, 1]$ is a symmetric function, referred to as the *graphon*, whose functional form is unknown to econometricians. It is therefore convenient to write A_{ij}

$$A_{ij} = f(X_i, \xi_i, X_j, \xi_j) + \epsilon_{ij}, \quad \forall 1 \leq i < j \leq N \quad (2.1)$$

where ϵ_{ij} are realization errors, which is conditionally independent across pairs $\{(i, j) \mid 1 \leq i < j \leq N\}$ given (X, ξ) .

The network formation model (2.1) is a conditionally independent dyadic model, because the graphon f depends only on the characteristics of the pair (i, j) and not on those of other individuals or links in the network. While A_{ij} and A_{ik} ($k \neq j$) may exhibit correlation unconditionally, they are independent given (X, ξ) . This setting is flexible enough to encompass a wide range of network formation models commonly used in economics and statistics. Below, we provide several examples.

Example 1 (Stochastic block model) *A widely used example consistent with our network formation framework is the stochastic block model (SBM). It assumes that each individual belongs to one of G groups, $g \in \{1, \dots, G\}$, and that the link formation probability depends only on group membership. In particular, the probability of a link between two individuals is determined solely by the groups to which they belong, so that individuals within the same group have the same connection probabilities toward other individuals. The stochastic block model corresponds to a special case of our network formation process in which the support of (X_i, ξ_i) is finite. Specifically, if (X_i, ξ_i) can take only G distinct values, the graphon f reduces to a piecewise constant function defined over G^2 group pairs.*

Example 2 (Network formation with transferable utility) Suppose that individuals i and j form a link if the total surplus from doing so is positive:

$$A_{ij} = \mathbf{1}(\omega(X_i, X_j)' \beta + g(\xi_i, \xi_j) - U_{ij} \geq 0), \quad (2.2)$$

where the total surplus consists of three components:

- **(Homophily in observables)** $\omega(X_i, X_j)$ measures the distance between observable characteristics of i and j . For instance, one may define $\omega(X_i, X_j) = (|X_{i1} - X_{j1}|, \dots, |X_{id_X} - X_{jd_X}|)'$ for absolute differences, or $\omega(X_i, X_j) = ((X_{i1} - X_{j1})^2, \dots, (X_{id_X} - X_{jd_X})^2)'$ for squared differences. A negative coefficient $\beta < 0$ captures homophily, meaning that individuals with similar observables are more likely to form a link.
- **(Individual heterogeneity)** The function $g(\xi_i, \xi_j)$ captures the influence of individual heterogeneity. For example, one may specify $g(\xi_i, \xi_j) = \xi_{i1} + \xi_{j1} - (\xi_{i2} - \xi_{j2})^2$, where the additive component $\xi_{i1} + \xi_{j1}$ represents degree heterogeneity (higher ξ_i implies more connections), while $-(\xi_{i2} - \xi_{j2})^2$ captures homophily in unobservables.
- **(Idiosyncratic shocks)** $\{U_{ij} \mid 1 \leq i < j \leq N\}$ captures the idiosyncratic preference shock that is independent of (X, ξ) . Let $F(\cdot)$ be the cumulative distribution of U_{ij} , then $\mathbb{P}(A_{ij} = 1 \mid X, \xi) = F(\omega(X_i, X_j)' \beta + g(\xi_i, \xi_j))$. Specifically, when U_{ij} follows type-I extreme value distribution, we have

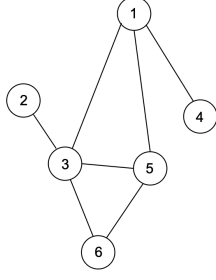
$$f(X_i, \xi_i, X_j, \xi_j) = \frac{\exp(\omega(X_i, X_j)' \beta + g(\xi_i, \xi_j))}{1 + \exp(\omega(X_i, X_j)' \beta + g(\xi_i, \xi_j))}$$

When $g(\xi_i, \xi_j)$ includes only the additive component $\xi_i + \xi_j$, the model reduces to the model of [Graham \(2017\)](#), which has been widely used in empirical applications.

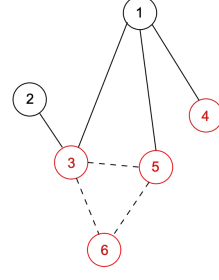
Beyond these examples, our network formation framework in (2.1) can flexibly incorporate richer forms of heterogeneity. For example, it allows for heterogeneous slope coefficients when the homophily effect on observables, β in (2.2), depends on the latent factors (ξ_i, ξ_j) . Similarly, the unobserved heterogeneity term $g(\xi_i, \xi_j)$ can capture more complex interaction structures, such as interactive fixed effects $\xi_i' \xi_j$. However, our dyadic model excludes strategic network formation, where the formation of one link may depend on the presence of other links (see [de Paula \(2020\)](#)).

2.2 Data

Economists observe individual covariates X_i (e.g., gender, age) for *all* i in the network but do not observe the full network A ; instead, only a sampled network is available. In this paper, we focus on egocentric sampling, which is one of the most commonly sampling designs in empirical work. Specifically, we randomly sample $n < N$ nodes in the network and ask all their friends not only in the sample but in the whole network. Let $\mathcal{S} \subset \{1, 2, \dots, N\}$ denote the set of sampled individuals with $|\mathcal{S}| = n$. Then a link A_{ij} is observed if $i \in \mathcal{S}$ or $j \in \mathcal{S}$, whereas links between two non-sampled nodes (i.e., $i \notin \mathcal{S}$ and $j \notin \mathcal{S}$) are unobserved. A simple example follows to illustrate the setup.



(a) Full network



(b) Sampled network by sampling $\{1, 2\}$, red nodes are not in the sample, dashed edges are not observed

$$\begin{pmatrix} 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ \hline 1 & 1 & \times & \times & \times & \times \\ 1 & 0 & \times & \times & \times & \times \\ 1 & 0 & \times & \times & \times & \times \\ 0 & 0 & \times & \times & \times & \times \end{pmatrix}$$

(c) From full adjacency matrix to partially observed adjacency matrix, $\times$ denotes missing entry

Figure 1: An example of how egocentric sampling works

Since it causes no harm to relabel the agents, we can do so such that $\mathcal{S} = \{1, 2, \dots, n\}$, in which case the observed adjacency matrix A^{obs} admits the following block structure:

$$A^{\text{obs}} = \begin{pmatrix} A_{\mathcal{S}\mathcal{S}} & A_{\mathcal{S}\mathcal{S}^c} \\ A_{\mathcal{S}^c\mathcal{S}} & \textcolor{red}{A_{\mathcal{S}^c\mathcal{S}^c}} \end{pmatrix} \quad (2.3)$$

Here, $A_{\mathcal{S}\mathcal{S}}$ consists of links between two sampled individuals, $A_{\mathcal{S}^c\mathcal{S}} = A'_{\mathcal{S}\mathcal{S}^c}$ contains links between sampled and unsampled individuals, and the bottom right block $A_{\mathcal{S}^c\mathcal{S}^c}$ is unobserved because both individuals are unsampled; we highlight this block in red.

2.3 Imputation

To illustrate the main idea of our imputation approach, let us start with a simple case in which both X_i and ξ_i are observable for every individual in the network. In this case, imputing missing links would be straightforward: we could estimate $f(\cdot, \cdot)$ nonparametrically from the observed adjacency matrix to obtain estimates $\hat{f}(\cdot, \cdot)$, and then use these estimates to impute the missing links in the submatrix $A_{\mathcal{S}^c\mathcal{S}^c}$.

The main difficulty, however, is that the latent factors ξ are unobserved for all individuals. To clarify the main challenge, let us first abstract from the observed covariates X and focus on the case where link formation depends solely on the latent factors. In this setting, the imputation problem reduces to recovering information about ξ from the observed subnetwork in order to predict the missing part of the adjacency matrix. Our approach is motivated by the following idea. When the network formation is dyadic as in (2.1), individuals with similar ξ are expected to have similar rows in the (denoised) adjacency matrix. This observation provides a useful heuristic: similarity in observed connections can indicate similarity in latent positions. Based on this idea, for any unsampled individual i , we find a set

of sampled individuals whose latent factors are close to ξ_i by comparing their connection patterns in the observed subnetwork. We then use the observed connections of these sampled individuals, weighted by their similarity to i , to impute the missing links for i .

Since the recovered information about ξ from the network is noisy whereas X is accurately measured, it is inappropriate to ignore the observed covariates or absorb them into the latent factors, as doing so would discard the noiseless information contained in X . Therefore, when X is available, our method first extracts the component of link formation explained by the observed covariates and then applies the latent-factor-based procedure described above to the residuals not captured by X . We finally combine these two components to obtain the final imputation of the network.

2.3.1 Extract information explained by X

For notational simplicity, we write $f_{ij} = f(X_i, \xi_i, X_j, \xi_j)$. As discussed previously, when individual covariates X are available, it is appropriate to first extract the component that can be explained by X . Consider the decomposition:

$$f_{ij} = \underbrace{\mathbb{E}(f_{ij} \mid X_i, X_j)}_{\Pi_{ij}} + \underbrace{f_{ij} - \mathbb{E}(f_{ij} \mid X_i, X_j)}_{\Pi_{ij}^\perp}$$

where Π_{ij} represents the part of the graphon that can be explained by the observed covariates, and $\Pi_{ij}^\perp$ captures the residual component of the graphon that cannot be explained solely by the observed covariates. Note that the observed link A_{ij} can then be written as $A_{ij} = \Pi_{ij} + (\Pi_{ij}^\perp + \epsilon_{ij})$, and the composite error term satisfies $\mathbb{E}(\Pi_{ij}^\perp + \epsilon_{ij} \mid X_i, X_j) = 0$, which implies that dyadic nonparametric regression of A_{ij} on (X_i, X_j) consistently estimates Π_{ij} . [Graham, Niu and Powell \(2021\)](#) derive the pointwise and uniform minimax risks for estimating such conditional expectations and show that the Nadaraya-Watson (NW) estimator attains the optimal convergence rate when using higher-order kernels¹. In addition, dyadic nonparametric regression can be implemented using observed subset of adjacency matrix, and since the covariates X are observed for every individual in the network, the nonparametric estimator $\hat{\Pi}_{ij}$ can be computed even when both i and j are unsampled. The implementation details are discussed in the Appendix.

In the next two steps, we focus on estimating this residual component $\Pi^\perp$ using pseudo-distance and two-way fixed-effect imputation procedure described below.

2.3.2 Calculate similarity

We now describe how to infer the similarity between latent factors using observed network links.

For notational simplicity, let $\zeta_i := (X_i, \xi_i)$ for each i , and let $f_{ij} = f(\zeta_i, \zeta_j)$ for all $1 \leq i, j \leq N$. Collect the elements f_{ij} into a matrix $F \in \mathbb{R}^{N \times N}$, and the elements ϵ_{ij} into a matrix $E \in \mathbb{R}^{N \times N}$. Then the adjacency matrix A can be expressed as $A = F + E$ ². Our approach relies on the following idea. For any unsampled individual i and sampled individual i' , if their denoised rows in observed network (or equivalently, the corresponding rows in F) appear sufficiently similar, then under suitable regularity conditions, ζ_i and $\zeta_{i'}$ are also similar in the latent space.

¹In practice, however, we use local polynomial regression rather than higher-order kernels, as it is more commonly used in local regression.

²Strictly speaking, the representation above is slightly imprecise. Because self-loops are ruled out, the diagonal entries of F are set to zero rather than f_{ii} . Likewise, we impose $\epsilon_{ii} = 0$ for each i .

Following [Zhang, Levina and Zhu \(2017\)](#), we define the pseudo-distance between i and i' as

$$\hat{d}_{ii'} = \max_{k=1,\dots,n} \left| \frac{1}{n} \sum_{\ell=1}^n A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|, \quad i, i' = 1, 2, \dots, N$$

The definition above relies only on the friendships of sampled individuals and can therefore be computed from the observed subnetwork. We will show that the pseudo-distance $\hat{d}$ effectively de-noises the adjacency matrix A , so that the distances computed from A and from the noise-free matrix F are asymptotically equivalent. In addition, under regularity conditions discussed later, a small value of $\hat{d}_{ii'}$ implies that the corresponding latent characteristics ζ_i and $\zeta_{i'}$ are sufficiently close in the latent space.

It is important to note that the discussion here concerns the composite characteristic ζ_i , rather than the observable covariate X_i alone. This is because (i) X_i and ξ_i enter the network formation process in a non-separable way, making it impossible to isolate their effects without additional assumptions; and (ii) the information in X_i that can be directly explained has already been extracted in the first step, so there is no need to distinguish between X_i and ξ_i in this step.

2.3.3 Local two-way fixed-effects imputation

We employ the pseudo-distance $\hat{d}$ obtained in the previous step to construct a local estimator for $\Pi_{ij}^\perp$, referred to as the two-way fixed-effect imputation. Specifically, for unsampled individuals i and j , we estimate the local two-way fixed-effects regression to obtain $\hat{\alpha}_i$ and $\hat{\alpha}_j$:

$$\hat{\alpha} \in \underset{\alpha \in \mathbb{R}^{n+2}}{\operatorname{argmin}} \sum_{\substack{i' \in \mathcal{S} \cup \{i\}, j' \in \mathcal{S} \cup \{j\} \\ (i', j') \neq (i, j)}} \left(A_{i'j'} - \hat{\Pi}_{i'j'} - \alpha_{i'} - \alpha_{j'} \right)^2 K_h(\hat{d}_{ii'}) K_h(\hat{d}_{jj'}) \quad (2.4)$$

where $K_h(\cdot)$ is a kernel function with bandwidth h , and $\hat{d}$ denotes the pseudo-distance. Then the imputed $\Pi_{ij}^\perp$ is given by $\hat{\Pi}_{ij}^\perp = \hat{\alpha}_i + \hat{\alpha}_j$, and the imputed network link is given by:

$$\hat{A}_{ij} = \hat{\Pi}_{ij} + \hat{\Pi}_{ij}^\perp = \hat{\Pi}_{ij} + \hat{\alpha}_i + \hat{\alpha}_j$$

The imputation is “local” in the sense that, by using a kernel estimator, we rely only on information from individuals i' and j' whose pseudo-distances $\hat{d}_{ii'}$ and $\hat{d}_{jj'}$ (or, equivalently, the distances between their latent positions) are sufficiently small when estimating fixed effects. Note that, although our notations omit the indices (i, j) for simplicity, the estimator $\hat{\alpha}$ in (2.4) is pair-specific and therefore must be re-estimated for each (i, j) .

The procedure is referred to as a “two-way fixed-effects imputation” because (2.4) takes the form of the two-way fixed-effects regression rather than a local constant regression. This is analogous to replacing a local constant estimator with a local polynomial estimator in nonparametric regression: incorporating the two-way fixed effects provides a higher-order local approximation and thus improves estimation accuracy. We will formally justify this result later using a Taylor expansion.

It is important to note that $\hat{A}_{ij}$ can only approximate the conditional link-formation probability f_{ij} rather than the missing link A_{ij} . This is because the missing link contains an unpredictable idiosyncratic error ϵ_{ij} , which cannot be predicted from the available data.

We impute all missing links based on the previous steps and merge these imputed entries with the observed ones to obtain a complete imputed matrix $\hat{A}$. To facilitate understanding of the main idea of our algorithm, we summarize below a simplified version of our method that excludes cross-fitting and cross-validation.

Algorithm 1 Imputation (Basic Idea)

Input: Kernel $K(\cdot)$, bandwidths h , incomplete network A^{obs} , covariate matrix X .

Output: Imputed network $\hat{A}$.

Step 1: Extract information explained by X

Run dyadic nonparametric regression to obtain $\hat{\Pi}_X(\cdot, \cdot)$.

Step 2: Calculate similarity

For each $i \in \mathcal{S}^c$ and $i' \in \mathcal{S}$, calculate the pseudo-distance using observed adjacency matrix:

$$\hat{d}_{ii'} = \max_{k=1, \dots, n} \left| \frac{1}{n} \sum_{\ell=1}^n A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|$$

Step 3: Local imputation

For each missing entry (i, j) , i.e., $i \in \mathcal{S}^c$ and $j \in \mathcal{S}^c$, run the local (weighted) two-way fixed effect regression

$$\hat{\alpha} \in \underset{\alpha \in \mathbb{R}^{n+2}}{\text{argmin}} \sum_{\substack{i' \in \mathcal{S} \cup \{i\}, j' \in \mathcal{S} \cup \{j\}, \\ (i', j') \neq (i, j)}} \left(A_{i'j'} - \hat{\Pi}_{i'j'} - \alpha_{i'} - \alpha_{j'} \right)^2 K_h(\hat{d}_{ii'}) K_h(\hat{d}_{jj'})$$

to obtain $\tilde{A}_{ij} = \hat{\alpha}_i + \hat{\alpha}_j + \hat{\Pi}_{X,ij}$. By combining the imputed entries with the observed entries of the adjacency matrix, we obtain the (preliminary) imputed adjacency matrix $\tilde{A}$. Truncate $\tilde{A}_{ij}$ to get

$$\hat{A}_{ij} = \begin{cases} \tilde{A}_{ij}, & \text{if } \tilde{A}_{ij} \in [0, 1] \\ 1 & \text{if } \tilde{A}_{ij} > 1 \\ 0 & \text{if } \tilde{A}_{ij} < 0 \end{cases}$$

and zero the diagonal of $\hat{A}$ such that $\hat{A}_{ii} = 0$ for all i .

2.3.4 Cross-validation and sample-splitting

Although the proposed algorithm is straightforward to implement, deriving the asymptotic properties of $\hat{A}_{ij}$ can be technically challenging because of the potential dependence between the estimated pseudo-distance $\hat{d}$ and the error terms ϵ in the two-way fixed-effects regression.

A standard approach to eliminate this dependence is to employ cross-fitting, in which the data are partitioned into folds so that the first-step and second-step estimations are conducted on separate subsets of the data. Specifically, we split the sample $\mathcal{S}$ into two parts, $\mathcal{S}_1$ and $\mathcal{S}_2$, of equal size. We then use only the friendships of $\mathcal{S}_1$ to calculate the similarity $\hat{d}(\cdot, \cdot)$:

$$\hat{d}_{ii'} = \max_{k=1, \dots, N} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|, \quad i \in \mathcal{S}_2 \cup \mathcal{S}^c, i' \in \mathcal{S}_2$$

In the imputation stage, we use no link information from $\mathcal{S}_1$ and all calculations rely on the friendships of $\mathcal{S}_2$.

$$\hat{\alpha} \in \underset{\alpha \in \mathbb{R}^{|\mathcal{S}_2|+2}}{\text{argmin}} \sum_{\substack{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}, \\ (i', j') \neq (i, j)}} \left(A_{i'j'} - \hat{\Pi}_{i'j'} - \alpha_{i'} - \alpha_{j'} \right)^2 K_h(\hat{d}_{ii'}) K_h(\hat{d}_{jj'})$$
$$\hat{A}_{ij} = \hat{\alpha}_i + \hat{\alpha}_j + \hat{\Pi}_{ij}$$

In addition, a practical concern is the choice of the tuning parameter h . We recommend selecting h using a leave-one-out cross-validation procedure, and the details of cross-validation, together with the sample-splitting, are provided in the Appendix.

3 Theory for Imputation

Our method accommodates both low-rank graphon and high-rank graphon, where the support of ζ is a continuum and f is nonlinear. In what follows, we focus on the second case, while noting that the same arguments apply, with minor modifications, to the low-rank graphon. We now introduce the regularity conditions.

Assumption 1 (Regularity Conditions) *Suppose that*

- (i) **(Asymptotics)** *Large- N , large- n asymptotics with $N, n \rightarrow \infty$.*
- (ii) **(Independence)** *$\{(X_i, \xi_i)\}_{i=1}^N$ are i.i.d., and conditional on (X, ξ) , $\{A_{ij}\}_{1 \leq i < j \leq N}$ are generated independently according to (2.1).*
- (iii) **(Random sampling)** *Let $D_i \in \{0, 1\}$ indicate whether i is sampled. Assume $D_i \perp (A, X, \xi)$ for all i , and $\{D_i\}_{i=1}^N$ are independent across i .*
- (iv) **(Compact support)** *Let $\zeta_i = (X'_i, \xi'_i)'$ and $d_\zeta = d_X + d_\xi$, assume that the support of ζ_i is compact on $\mathbb{R}^{d_\zeta}$. In addition, there exist constants $0 < \underline{c} < \bar{c} < \infty$ such that for any $\zeta \in \text{supp}(\zeta)$ and any $h > 0$, $\underline{c}h^{d_\zeta} \leq \mathbb{P}(\|\zeta_i - \zeta\| \leq h) \leq \bar{c}h^{d_\zeta}$.*
- (v) **(Smoothness)** *For every fixed $\zeta_j \in \text{supp}(\zeta)$, the mapping $\zeta \mapsto f(\zeta, \zeta_j)$ is twice continuously differentiable, its second derivative is uniformly bounded for all $\zeta, \zeta_j \in \text{supp}(\zeta)$.*

Assumption 1(i) indicates that our method is designed for large-network settings in which both the network size N and the number of sampled individuals n grow large. Our analysis does not rely on n and N growing at the same rate and continues to hold when n grows more slowly, e.g., $n = O(N^\alpha)$ for any $\alpha \in (0, 1)$.

Assumption 1(ii) imposes that $\{(X_i, \xi_i)\}_{i=1}^N$ are i.i.d., and the data-generating process of A follows a conditionally independent dyadic model (2.1). This model also follows from the Aldous-Hoover Theorem (Aldous, 1981; Hoover, 1979), which shows that any infinitely exchangeable random graph admits such a representation. As discussed earlier, although (2.1) is flexible enough to encompass a wide range of network formation models commonly used in economics and statistics, it rules out strategic network formation, in which individuals linking decisions depend on those of others; see de Paula (2020) for a review.

Assumption 1(iii) is standard in the egocentric sampling design, which assumes that individuals are randomly drawn from the population independently of (A, X, ξ) . However, this assumption can be relaxed; for example, D_i may depend arbitrarily on (X_i, ξ_i) as long as $\epsilon_{ij} \perp (D_i, D_j)$ conditional on (X_i, ξ_i, X_j, ξ_j) .

The first part of Assumption 1(iv) imposes the boundedness of ζ , and the second part introduces a standard bounded-density condition. Intuitively, it ensures that the distribution of ζ_i is neither locally too sparse nor too concentrated by requiring that, for any point ζ and a small radius $h > 0$, the probability that ζ_i falls within a ball of radius h centered at ζ is of the same order as the volume of that ball. This requirement is automatically satisfied when the support is convex and the probability density function of ζ is uniformly bounded above and bounded away from zero. This assumption is common for establishing the asymptotic properties of nonparametric kernel estimators.

Assumption 1(v) requires the graphon f to be twice continuously differentiable with uniformly bounded second derivatives. This smoothness condition is needed because our two-way fixed-effects imputation method relies on a second-order Taylor expansion of $\Pi_{ij}^\perp$ within a local neighborhood to eliminate the first-order approximation bias, thereby improving accuracy. Methods that rely on higher-order Taylor expansions, such as the local PCA approach proposed by Feng (2023), correspondingly require the existence of higher-order derivatives. We discuss this point in detail later.

It should be noted that the requirement of second-order differentiability of f is not a mild condition. For instance, consider the network formation model with degree heterogeneity in Example 2. If the distance function $\omega(\cdot, \cdot)$ is defined by the absolute distance, i.e., $\omega(X_i, X_j) = |X_i - X_j|$, which is common in applied research, then the corresponding graphon f is at most Lipschitz continuous rather than twice differentiable. Nevertheless, our method remains valid and can consistently impute the conditional probability of link formation, although the approximation error no longer attains the second-order rate.

3.1 Asymptotic analysis of pseudo-distance

We now develop the asymptotic properties of the pseudo-distance $\hat{d}_{ii'}$ and show that a sufficiently small $\hat{d}_{ii'}$ implies that ζ_i and $\zeta_{i'}$ are close in the latent space. In the following lemma, we establish the convergence of $\hat{d}_{ii'}$ to its population counterpart $d_{ii'}$, defined as

$$d_{ii'} = \sup_{\zeta_k \in \text{supp}(\zeta)} \left| \int_{\zeta_\ell} g(\zeta_k, \zeta_\ell) (g(\zeta_i, \zeta_\ell) - g(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right|$$

Lemma 1 *Under Assumption 1, there exist constants $\gamma_1, \gamma_2 > 0$ such that*

$$\mathbb{P} \left(\max_{i \in \mathcal{S}_2 \cup \mathcal{S}_2^c, i' \in \mathcal{S}_2} |\hat{d}_{ii'} - d_{ii'}| \geq \gamma_1 \frac{\log N}{N^{1/d_\zeta}} + \gamma_2 \frac{\log N}{\sqrt{n}} \right) \leq 1 - n^{-\frac{1}{2}}$$

Lemma 1 shows that the approximation error of $\hat{d}_{ii'}$ relative to its population counterpart $d_{ii'}$ can be decomposed into two components. The first term, $\gamma_1 \frac{\log N}{N^{1/d_\zeta}}$, captures the matching discrepancy in the latent characteristics ζ_i . Its rate deteriorates with the dimension of the latent space d_ζ , which is consistent with the well-known “curse of dimensionality” in the matching literature. The second term, $\gamma_2 \frac{\log N}{\sqrt{n}}$, reflects the effect of the idiosyncratic error ϵ_{ij} . It shows that although the true graphon is unobservable, the pseudo-distance can asymptotically denoise the observed network. The matching discrepancy error dominates the effect of idiosyncratic errors when $N \sim n$ and $d_\zeta \geq 3$, which is the common case in practice. Finally, for notational simplicity, let

$$\delta_{N,n} := \gamma_1 \frac{\log N}{N^{1/d_\zeta}} + \gamma_2 \frac{\log N}{\sqrt{n}}$$

and henceforth focus on the event under which $|\hat{d}_{ii'} - d_{ii'}|$ is uniformly bounded by $\delta_{N,n}$, as implied by Lemma 1.

The following assumption is critical and requires that the population pseudo-distance $d_{ii'}$ be sufficiently informative to reveal the latent distance $\|\zeta_i - \zeta_{i'}\|$.

Assumption 2 (Informativeness) *For all $\zeta_i, \zeta_{i'} \in \text{supp}(\zeta)$, there exists a constant $\lambda > 0$ such that $d_{ii'} \geq \lambda \|\zeta_i - \zeta_{i'}\|$.*

This assumption ensures that when the population pseudo-distance $d_{ii'}$ is sufficiently small, the latent positions ζ_i and $\zeta_{i'}$ are close to each other. Hence, $d_{ii'}$ can be used to identify individuals with similar latent characteristics.

Example 1 (Continued) When each individual belongs to one of G groups, $g \in \{1, \dots, G\}$, Assumption 2 is guaranteed as long as there is heterogeneity across different groups, that is, no two groups share identical linking patterns. A formal illustration and the proof are provided in Lemma D.1 in the Appendix. When the graphon f takes a more general low-rank form, the informativeness condition still holds under similarly mild assumptions. We refer the reader to the discussions in Feng (2023) and Deaner, Hsiang and Zelenev (2025) for further details.

Example 2 (Continued) In Example 2, the latent type ζ may lie on a continuous support, and the graphon f is not necessarily low rank. Therefore, verifying Assumption 2 in this setting is generally challenging. Nevertheless, we show that the informativeness assumption holds in the following model which serves as a benchmark in empirical applications. Specifically, Assumption 2 holds when (i) U_{ij} in (2.2) follows a Type-I extreme value distribution, and (ii) the model allows for degree heterogeneity and homophily effects on both observable and unobservable characteristics, where homophily can be captured by either absolute or quadratic distance. This finding implies that the informativeness assumption is empirically plausible and unlikely to impose restrictive constraints in applied network analysis. A formal illustration and the proof are provided in Lemma D.2 in the Appendix.

Remark An alternative to the pseudo-distance is the Euclidean distance. More precisely, for any pair of nodes i and i' , define the squared Euclidean distance as $\hat{r}_{ii'}^2 = \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} (A_{i\ell} - A_{i'\ell})^2$. The main drawback of using the squared Euclidean distance is that it does not automatically denoise the observed network, because $\hat{r}_{ii'}^2$ includes the inner products of each error vector with itself, $\frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} \epsilon_{i\ell}^2 + \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} \epsilon_{i'\ell}^2$, which do not vanish even as $n \rightarrow \infty$. In contrast, the pseudo-distance involves cross inner products of independent error vectors, so that the influence of noise decreases at the $1/\sqrt{n}$ rate, thereby asymptotically denoising the adjacency matrix.

Another alternative to the pseudo-distance is the difference in average degrees, defined as $\hat{\rho}_{ii'} = \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} A_{i\ell} - \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} A_{i'\ell}$. Although the average-degree distance can also denoise the adjacency matrix, it requires a stronger informativeness assumption than Assumption 2. In particular, using average-degree distance requires the population average degree,

$$\zeta \mapsto \int_{\zeta'} f(\zeta, \zeta') \mathbb{P}(\zeta')$$

be a strictly monotonic function of ζ . This monotonicity requirement corresponds to the injectivity condition in Bonhomme, Lamadon and Manresa (2022). However, the condition does not hold in either Example 1 or Example 2 and is stronger than the Assumption 2 in our paper.

3.2 Asymptotic analysis of imputation

Since the imputation procedure involves two parts, a nonparametric dyadic regression and a two-way fixed-effects regression, the imputation error can accordingly be decomposed into two parts. Our asymptotic analysis focuses on the second part, as the first-stage nonparametric estimator is generally much more accurate. The asymptotic properties of the first-stage estimator have been well established in the literature, see Graham, Niu and Powell (2021).

Assumption 3 The estimation error of $\hat{\Pi}_{ij}$ admits the following uniform bound: there exist constants $\gamma_3, \gamma_4 > 0$ such that

$$\sup_{1 \leq i, j \leq N} \mathbb{E} \left(\hat{\Pi}_{ij} - \Pi_{ij} \right) \leq \gamma_3 n^{\frac{-2}{4+d_X}} \log(N), \quad \sup_{1 \leq i, j \leq N} \mathbb{V}^{1/2} \left(\hat{\Pi}_{ij} \right) \leq \gamma_4 n^{\frac{-2}{4+d_X}} \log^2(N)$$

Assumption 3 states that the first-stage nonparametric dyadic regression estimator admits uniform bounds on its bias and standard deviation. Up to logarithmic factors, both bounds attain the rate $n^{-2/(4+d_X)}$, which is the optimal nonparametric mean-squared-error rate for a twice continuously differentiable target function with d_X -dimensional covariates. Under Assumption 1, [Graham, Niu and Powell \(2021\)](#) show that the Nadaraya-Watson estimator with a higher-order kernel satisfies Assumption 3.

Remark First, although the dyadic conditional expectation $\Pi(\cdot, \cdot)$ takes $2d_X$ arguments, the nonparametric convergence rate in Assumption 3 behaves as if Π depended on only d_X arguments, making it consistent with the conclusion in [Graham, Niu and Powell \(2021\)](#). Second, as noted by [Graham, Niu and Powell \(2021\)](#), when the entire network is observed, the regression uses $O(N^2)$ dyads, but the effective sample size is just $O(N)$ because the estimator is a U-statistic. In our settings, however, the effective sample size is of order n (not $\sqrt{Nn}$) as stated in Assumption 3. This argument can be verified via Hoeffding decomposition of dyadic nonparametric estimators.

Assumption 4 Suppose that the kernel $K : \mathbb{R} \mapsto \mathbb{R}_+$ is bounded and supported on $[-1, 1]$. In addition, $K(0) > 0$ and $K(\cdot)$ is Lipschitz continuous with constant $\bar{K} > 0$.

This assumption imposes regularity conditions on kernel K used in the second step. The requirement is standard in the literature, and many commonly used kernels, including the Epanechnikov kernel, satisfy Assumption 4.

We now briefly explain why $\hat{d}_{ii'}$ can be viewed as a noisy measure of $\|\zeta_i - \zeta_{i'}\|$ and can thus serve as its proxy in the kernel function. First, let h denote the bandwidth of the kernel function. When imputing $\Pi_{ij}^\perp$, Assumption 4 allows us to restrict attention to links (i', j') for which both $\hat{d}_{ii'} \leq h$ and $\hat{d}_{jj'} \leq h$. Second, by Lemma 1, we have $|\hat{d}_{ii'} - d_{ii'}| \leq \delta_{N,n}$ with high probability. When $\delta_{N,n}/h \rightarrow 0$, the approximation error in $\hat{d}_{ii'}$ is asymptotically negligible relative to the bandwidth h , so $d_{ii'}$ can be replaced by $\hat{d}_{ii'}$ without loss of first-order accuracy. Furthermore, under Assumption 2, $d_{ii'}$ is informative about the latent distance in the sense that $\|\zeta_i - \zeta_{i'}\| \leq \lambda^{-1} d_{ii'}$. Combining these results yields, with probability approaching one, $\|\zeta_i - \zeta_{i'}\| \leq \lambda^{-1} h$, which implies that using pseudo-distances in the kernel function selects pairs whose latent distances lie within a $\lambda^{-1} h$ -neighborhood when imputing missing links.

The following theorem provides the theoretical guarantee for our imputation method.

Theorem 2 (Imputation errors) Under Assumption 1, 2, 3, 4, if $\delta_{N,n}/h \rightarrow 0$, $nh^{d_\zeta} \rightarrow \infty$, then

$$\max_{i,j \in \mathcal{S}^c} \mathbb{E}(\hat{A}_{ij} - f_{ij}) = O(h^2), \quad \max_{i,j \in \mathcal{S}^c} \mathbb{V}^{1/2}(\hat{A}_{ij}) = O\left(\frac{\log N}{\sqrt{nh^{d_\zeta}}} + \delta_{N,n}h\right)$$

Theorem 2 establishes entry-wise uniform error bounds, rather than a Frobenius-norm bound for a submatrix. In contrast to the graphon estimation literature (e.g., [Gao, Lu and Zhou \(2015\)](#) and [Xu \(2018\)](#)), our result provides guarantees for link-level prediction of the underlying graphon.

Following the classical nonparametric regression literature, we decompose the estimation error into bias and variance components. The bias term is of order h^2 because our local two-way fixed-effects regression removes the first-order bias, leaving only the second-order approximation error. The variance consists of two parts. The first term, $(\log N)^2/(nh^{d_\zeta})$, matches the rate in the standard nonparametric literature. The second term, $\delta_{N,n}h$, is special to our estimator and arises from the use of the pseudo-distance as a substitute for the unobserved latent factors ζ_i and ζ_j . When the latent factors are observable, we have $\delta_{N,n} = 0$, and the second term disappears, reducing our result to cases [Tsybakov \(2008\)](#) and [Graham, Niu and Powell \(2021\)](#).

The bandwidth h governs the bias-variance trade-off. As h decreases, the bias diminishes; however, because our variance consists of two components, the trade-off deviates from the classical setting. While

$\frac{\log N}{\sqrt{nh^{d_\zeta}}}$ increases as h shrinks, the term $\delta_{N,n}h$ decreases. Therefore, the optimal choice of h depends jointly on the relative magnitudes of n and N and on d_ζ . However, when n and N are of the same order, one can still set $h \asymp n^{-1/(4+d_\zeta)}$ and attain the minimum mean-squared-error.

Remark Our method employs two-way fixed-effects regression for imputation, which effectively removes the first-order approximation error. The intuition behind this approximation can be illustrated by a Taylor expansion. Heuristically, to impute A_{ij} , if we ignore the first-step nonparametric dyadic regression and consider the Taylor expansion of $f(\zeta_{i'}, \zeta_{j'})$ around the fixed point (ζ_i, ζ_j) :

$$\begin{aligned} f(\zeta_{i'}, \zeta_{j'}) &= \underbrace{f(\zeta_i, \zeta_j) + (\nabla_{\zeta_i} f(\zeta_i, \zeta_j))' (\zeta_{i'} - \zeta_i)}_{\alpha_{i'}} + \underbrace{(\nabla_{\zeta_j} f(\zeta_i, \zeta_j))' (\zeta_{j'} - \zeta_j)}_{\alpha_{j'}} + O(\|\zeta_{i'} - \zeta_i\|^2 + \|\zeta_{j'} - \zeta_j\|^2) \\ &= \alpha_{i'} + \alpha_{j'} + O(\|\zeta_{i'} - \zeta_i\|^2 + \|\zeta_{j'} - \zeta_j\|^2) \end{aligned}$$

Because we employ a kernel-weighted estimator based on the pseudo-distances $\hat{d}_{ii'}, \hat{d}_{jj'}$ with bandwidth h , and since, as discussed above, $\hat{d}_{ii'}$ and $\hat{d}_{jj'}$ differ from the latent-space distances $\|\zeta_{i'} - \zeta_i\|$ and $\|\zeta_{j'} - \zeta_j\|$ only up to a bounded factor, the contribution to imputation comes from links (i', j') with $\hat{d}_{ii'}, \hat{d}_{jj'} \leq h$. For those links, we obtain

$$f(\zeta_{i'}, \zeta_{j'}) = \alpha_{i'} + \alpha_{j'} + O(h^2)$$

Hence, the two-way fixed-effects regression absorbs all first-order approximation errors, leaving only the second-order remainder $O(h^2)$.

The idea of using two-way fixed-effects imputation is closely related to the approaches developed in [Freeman and Weidner \(2023\)](#) and [Beyhum and Mugnier \(2024\)](#). Our approach differs from theirs in two respects. First, their methods do not involve missing data and are primarily designed to estimate structural parameters in panel data models, whereas our method aims to predict the formation probability of missing links. Second, they discretize the data into groups using k-means clustering, while we employ a kernel estimator.

The methods most conceptually related to ours are those of [Feng \(2023\)](#) and [Deaner, Hsiang and Zelenev \(2025\)](#). [Deaner, Hsiang and Zelenev \(2025\)](#) propose a local-constant estimator to impute potential outcomes. Because they assume only that the potential outcome function is Lipschitz continuous, their approximation error is of order $O(h)$. Building on the ideas of [Freeman and Weidner \(2023\)](#) and [Beyhum and Mugnier \(2024\)](#), we assume that the graphon is twice continuously differentiable and, by using a two-way fixed-effects imputation, we reduce the approximation error to $O(h^2)$. In comparison, the local PCA method proposed by [Feng \(2023\)](#) leverages factor models together with higher-order Taylor expansions to achieve more accurate approximations. In particular, [Feng \(2023\)](#) show that the local PCA estimator attains Stone's minimax-optimal rate for nonparametric regression. We refer interested readers to that paper for further details.

We adopt the two-way fixed-effects model instead of local PCA approach for three reasons. First, local PCA method requires prior knowledge of the dimension d_ζ , which is typically unknown in practice and rarely supported by prior information. In contrast, our approach does not require prior knowledge of d_ζ . Second, the higher-order smoothness assumption is often violated in empirically relevant settings. As discussed earlier, in many applied network formation models, the homophily effect depends on the absolute distance between characteristics, implying that the graphon is only Lipschitz continuous. Our approach is robust to such lack of smoothness and can still achieve first-order approximation accuracy in this case. Third, networks in most economic applications are relatively small. For instance, in development economics, village networks typically contain around 200 households, and with a sampling

rate of roughly 40%, the effective sample size available for imputing a specific missing link can be fewer than 20. Under such small-sample conditions, achieving theoretical higher-order accuracy is difficult. Our numerical simulations also show that the finite-sample performance of our method is better to that of the local PCA.

4 Theory for Downstream Estimation

We consider the application of our method to estimating parameters in downstream economic models. The data $\{(A_m, X_m, W_m, Y_m)\}_{m=1}^M$ consist of M large i.i.d. networks indexed by $m = 1, 2, \dots, M$, and each network has size N_m . Here, $A_m \in \{0, 1\}^{N_m \times N_m}$ is the adjacency matrix, $Y_m \in \mathbb{R}^{N_m}$ is the outcome variable, $X_m \in \mathbb{R}^{N_m \times d_X}$ denotes the covariates that directly enter the network formation model f as in (2.1), whereas $W_m \in \mathbb{R}^{N_m \times d_W}$ contains variables that do not necessarily appear in the network formation model but affect the outcome Y_m . Note that W_m and X_m may overlap, i.e., $W_m \cap X_m \neq \emptyset$.

The adjacency matrix A_m is not fully observed because of egocentric sampling, but the outcome Y_m and the covariates (X_m, W_m) are observed for all individuals in the network. Let n_m be the number of sampled individuals in network m . We mainly focus on the asymptotics in which $M, N_m, n_m \rightarrow \infty$. The large-network asymptotics $N_m, n_m \rightarrow \infty$ are required for consistent imputation of the adjacency matrix, but letting $M \rightarrow \infty$ is not necessary; our analysis also applies to singular large network.

4.1 Consistency

GMM We work with the moment function

$$\psi(A_m, X_m, W_m, Y_m, \alpha) : [0, 1]^{N_m \times N_m} \times \mathbb{R}^{N_m \times d_X} \times \mathbb{R}^{N_m \times d_W} \times \mathbb{R}^{d_\alpha} \mapsto \mathbb{R}^q$$

where q is the number of moment conditions and $\alpha \in \mathbb{R}^{d_\alpha}$ is the structural parameter of interest in the downstream analysis. We assume the following moment condition:

$$\mathbb{E}(\psi(A_m, W_m, Y_m, \alpha_0)) = 0 \quad (4.1)$$

where α_0 is the true parameter. Since A_m is partially observed due to sampling, we impute each sampled network separately to obtain $\{\hat{A}_m \mid m = 1, \dots, M\}$, and then use the imputed network to solve:

$$\hat{\alpha} = \arg \min_{\alpha \in \text{supp}(\alpha)} \left(\frac{1}{M} \sum_{m=1}^M \psi(\hat{A}_m, W_m, Y_m, \alpha) \right)' \Sigma_M \left(\frac{1}{M} \sum_{m=1}^M \psi(\hat{A}_m, W_m, Y_m, \alpha) \right) \quad (4.2)$$

where $\Sigma_M \in \mathbb{R}^{q \times q}$ is the GMM weight matrix.

Example 3 (Regression on centrality) A large literature studies the relationship between individuals' network positions (centrality) and their outcomes, for example, [Hochberg, Ljungqvist and Lu \(2007\)](#), [Cruz, Labonne and Querubin \(2017\)](#), and [Banerjee et al. \(2013\)](#). The regressions used in such studies can be written as:

$$Y_{mi} = \alpha_C + \alpha_1 \phi_{mi}(A_m) + e_{mi}, \quad i = 1, 2, \dots, N_m, \quad m = 1, 2, \dots, M \quad (4.3)$$

where $\alpha = (\alpha_C, \alpha_1)'$ is the parameter of interest, $\phi_{mi} : [0, 1]^{N_m \times N_m} \rightarrow \mathbb{R}$ computes the centrality of individuals i . For example:

- (i) **(Degree centrality)** For any matrix $\tilde{A}_m \in [0, 1]^{N_m \times N_m}$, if $\phi_{mi}(\tilde{A}_m) = \frac{1}{N_m} \sum_{j=1}^{N_m} \tilde{A}_{m,ij}$, then $\phi_{mi}(\cdot)$ represents the normalized degree of individual i when $\tilde{A}$ is an adjacency matrix.

(ii) (**Eigenvector centrality**) For any symmetric matrix $\tilde{A}_m \in [0, 1]^{N_m \times N_m}$, let $\tilde{A}_m = U_m D_m U_m'$ be the eigenvalue decomposition of $\tilde{A}$, where $U_m \in \mathbb{R}^{N_m \times N_m}$ is an orthonormal matrix containing the eigenvectors of $\tilde{A}_m$ and $D_m \in \mathbb{R}^{N_m \times N_m}$ is a diagonal matrix with eigenvalues arranged in decreasing order. Define $\phi_{mi}(\tilde{A}_m) = \sqrt{N_m} U_{m,i1}$, where $U_{m,i1}$ denotes the i -th entry of the leading eigenvector. When $\tilde{A}_m$ is a symmetric adjacency matrix, $\phi_{mi}(\cdot)$ corresponds to the eigenvector centrality of individual i .

$\{e_{mi} \mid m = 1, \dots, M, i = 1, \dots, N_m\}$ contains the error terms that may be correlated within networks, so inference should be clustered at the network level. We use the moment function:

$$\psi(\hat{A}_m, W_m, Y_m, \alpha) := \frac{1}{N_m} \sum_{i=1}^{N_m} (1, \phi_{mi}(\hat{A}_m))' (Y_{mi} - \alpha_C - \alpha_1 \phi_{mi}(\hat{A}_m))$$

to obtain estimates of α . When the errors are independent within each network, our results extend to the case of a single large network (i.e., $M = 1$), and we still obtain a consistent estimator.

Example 4 (Linear-in-means peer-effects model) The linear-in-means peer-effects model is widely used to study peer influence in social networks. The model is:

$$Y_m = \alpha_C + \alpha_{\bar{Y}} G_m Y_m + \alpha_W W_m + \alpha_{\bar{W}} G_m W_m + e_m \quad (4.4)$$

where $\alpha = (\alpha_C, \alpha_{\bar{Y}}, \alpha'_W, \alpha'_{\bar{W}})'$, $\alpha_{\bar{Y}}$ captures the endogenous effect, and G_m is the row-normalized adjacency matrix³. $G_m Y_m$ is endogenous due to the reflection problem (Manski (1993)), the model must be estimated using instrumental variables. Under the identification conditions in Bramoullé, Djebbari and Fortin (2009), the following moment function can be constructed:

$$\psi(A_m, W_m, Y_m, \alpha) = \frac{1}{N_m} Z_m' (Y_m - \alpha_C - \alpha_{\bar{Y}} G_m Y_m - \alpha_W W_m - \alpha_{\bar{W}} G_m W_m)$$

where $Z_m = [1, W_m, G_m W_m, G_m^2 W_m]$ is the instrument. When the network A_m is incomplete, we compute $\hat{G}_m$ and $\hat{Z}_m$ using imputed network $\hat{A}$ ⁴, and use the moment function:

$$\psi(\hat{A}_m, W_m, Y_m, \alpha) = \frac{1}{N_m} \hat{Z}_m' (Y_m - \alpha_C - \alpha_{\bar{Y}} \hat{G}_m Y_m - \alpha_W W_m - \alpha_{\bar{W}} \hat{G}_m W_m)$$

to obtain α . Since $\{e_{mi} \mid m = 1, \dots, M, i = 1, \dots, N_m\}$ contains the error terms that may be correlated within networks, so inference should be clustered at the network level. However, when the errors are independent within each network, our results extend to the case of a single large network (i.e., $M = 1$), and we still obtain a consistent estimator.

Remark Our approach for estimating GMM parameters differs from that of Chandrasekhar and Lewis (2011). Specifically, Chandrasekhar and Lewis (2011) construct moments by integrating over the missing network data, typically using simulation to approximate (conditional) expectations. As a consequence,

³For any $i, j \in 1, \dots, N_m$, $G_{m,ij}$ is obtained through:

$$G_{m,ij} = \frac{A_{m,ij}}{\sum_{j'=1}^{N_m} A_{m,ij'}}$$

⁴For any $i, j \in 1, \dots, N_m$, $\hat{G}_{m,ij}$ is obtained through:

$$\hat{G}_{m,ij} = \frac{\hat{A}_{m,ij}}{\sum_{j'=1}^{N_m} \hat{A}_{m,ij'}}$$

and $\hat{Z}_m = [1, W_m, \hat{G}_m W_m, \hat{G}_m^2 W_m]$

their implementation is computationally demanding in GMM settings and relies on parametric assumptions about the distribution of the error terms (see the discussion in [Chandrasekhar and Lewis \(2011\)](#) and [Boucher and Houndetoungan \(2020\)](#)). By contrast, our approach directly plugs the imputed network into the moment function $\psi(\cdot)$, avoiding simulation and distributional assumptions on the errors. A key advantage of [Chandrasekhar and Lewis \(2011\)](#)'s approach is that it can accommodate complex network statistics, such as average path length, that cannot be computed using a simple plug-in strategy. However, even in this case, our imputed adjacency matrix can still serve as the basis for the simulation-based GMM procedure using the theoretical framework developed in Section 4.2 of [Chandrasekhar and Lewis \(2011\)](#).

Assumption 5 (GMM) *Suppose that*

- (i) **(Asymptotics)** *Large- M asymptotics with $M \rightarrow \infty$.*
- (ii) **(Independence)** *$\{(A_m, X_m, W_m, Y_m)\}_{m=1}^M$ are i.i.d. across networks.*
- (iii) **(Identification)** *There is a unique α_0 such that $\mathbb{E}(\psi(A_m, W_m, Y_m, \alpha_0)) = 0$.*
- (iv) **(Idiosyncrasy-robustness)** *Let F_m be the graphon matrix A_m , then as $N \rightarrow \infty$:*

$$\sup_{1 \leq m \leq M} \sup_{\alpha} \|\psi(F_m, W_m, Y_m, \alpha) - \psi(A_m, W_m, Y_m, \alpha)\| = o_P(1)$$
- (v) $\Sigma_M \xrightarrow{P} \Sigma$ *and Σ is positive definite.*
- (vi) **(Compactness and integrable)** *The support of α is compact, $\alpha \mapsto \psi(A_m, W_m, Y_m, \alpha)$ is continuous almost everywhere, and $\mathbb{E} \left(\sup_{\alpha \in \text{supp}(\alpha)} \|\psi(A_m, W_m, Y_m, \alpha)\|^2 \right) < \infty$.*
- (vii) *Let $\mathcal{S}_m$ be the set of sampled nodes in $\mathcal{A}$, then $\sup_{m \leq M} N_m^{-1} \|\hat{A}_{m, \mathcal{S}_m^c \mathcal{S}_m^c} - F_{m, \mathcal{S}_m^c \mathcal{S}_m^c}\|_F = o_P(1)$.*
- (viii) **(Lipschitz)** *For any $F_{m1}, F_{m2} \in [0, 1]^{N \times N}$, we have:*

$$\sup_{\alpha \in \text{supp}(\alpha)} \|\psi(F_{m1}, W_m, Y_m, \alpha) - \psi(F_{m2}, W_m, Y_m, \alpha)\| \leq L(W_m, Y_m) N_m^{-1} \|\hat{F}_{m1} - F_{m2}\|_F$$

$$\text{and } \mathbb{E} L^2(W_m, Y_m) < \infty.$$

Assumption 5(i) and (ii) are standard to ensure the consistency of M-estimator. However, if we are willing to assume that the error terms are independent within networks, then these two assumptions are no longer needed, and our analysis can apply to studying single large network; see the discussion in Examples 3 and 4.

Assumption 5(iii) is a standard identification condition and ensures that the moment condition does not hold at $\alpha \neq \alpha_0$. However, in the large-network asymptotic regime, weak identification becomes a concern. For example, in Example 4, if W is i.i.d. within each network and independent of the network (e.g., random treatment assignment), the instrument Z becomes asymptotically collinear, leading to an inconsistent estimator (see [Hayes and Levin \(2024\)](#) for further discussion). In the following analysis, we therefore assume that W is correlated with X to avoid the weak-identification problem.

Assumption 5(iv) implies that the difference between using graphon matrix F and the adjacency matrix A in the moment condition becomes asymptotically negligible as the network size N grows. This assumption is key for establishing the consistency of our plug-in estimator: because our imputation can only approximate graphon f instead of missing links, our estimator is consistent only if replacing A with F leaves the moment condition unchanged asymptotically. Since A decomposes into underlying

graphon matrix F , which is predictable from observed information, and an unpredictable error matrix, this assumption requires that the contribution of the unpredictable error matrix vanishes asymptotically. This condition is satisfied in many empirical applications using network data, such as the linear-in-means peer-effects model and linear regressions on network centralities, because network statistics aggregate a large number of links and naturally denoise link-level errors (see the discussion in the following text).

Assumption 5(v) and (vi) are standard conditions for establishing the consistency of M-estimators. Assumption 5(vii) states that, for each network m , the imputed network $\hat{A}_m$ is consistent for the graphon matrix F_m on the missing subnetwork in the Frobenius norm. This condition is weaker than the conclusion in Theorem 4, as the latter establishes consistency in the sup norm for the missing submatrix.

Assumption 5(viii) imposes a Lipschitz condition requiring that perturbations in F_m lead to changes in ψ up to a factor $L(W_m, Y_m)$ with a finite second moment, uniformly over α . Since our parameter of interest is α and F_m serves as a nuisance parameter, this assumption can be interpreted as a smoothness condition on the nuisance parameters within the framework of semi-parametric M-estimation.

Example 3 (Continued) *We discuss key conditions, Assumption 5(iv) and (viii), for linear regressions on degree centrality and eigenvector centrality, respectively. For degree centrality, verification is straightforward. Because degree centrality is simply the row average of the adjacency matrix, the difference between moments constructed from the observed adjacency matrix A_m and the underlying graphon matrix F_m is proportional to the row average of error terms. Since the row average of errors converges at rate $O_p(1/\sqrt{N_m})$, Assumption 5 follows immediately. A similar argument, together with the Cauchy-Schwarz inequality, verifies Assumption 5(viii).*

For eigenvector centrality, we impose a standard eigen-gap condition requiring that the largest eigenvalue of F_m is separated from the second-largest eigenvalue uniformly across networks. Such conditions are common in spectral analysis and ensure that the leading eigenspace is well separated. Under this assumption, the Davis-Kahan theorem Yu, Wang and Samworth (2015) implies that the perturbation in eigenvector centrality is proportional to the normalized spectral norm of the error matrix, i.e., $\|E\|_{\text{op}}/N_m$, which is of order $1/\sqrt{N_m}$ by Bandeira and Van Handel (2016). This establishes Assumption 5(iv). Assumption 5(viii) follows by a similar argument.

Example 4 (Continued) *For the linear-in-means peer-effects model, if each element in W_m is uniformly bounded and the strict exogeneity condition $\mathbb{E}(e_{m,i} \mid W_m) = 0$ holds for every i , then Assumptions 5(iv) and (viii) can be verified by arguments analogous to those used for the centrality regressions.*

Theorem 3 (Consistency of estimator) *Under Assumption 5, the estimator $\hat{\alpha}$ defined as in (4.2) is consistent for α_0 .*

Theorem 3 is particularly relevant in empirical applications, where interest typically focuses on the consistency of parameter estimates constructed from the imputed network rather than on the accuracy of the imputed links themselves. This theorem shows that the plug-in estimator based on our imputed adjacency matrices is consistent. The key advantages of our approach are that it is robust under misspecification of (i) the network formation model, (ii) the dimension of latent heterogeneity, and (iii) the smoothness of the network formation process. Moreover, even when a researcher prefers to impose a specific parametric network formation model, for computational simplicity or due to strong prior knowledge, our nonparametric procedure still provides a useful benchmark for assessing how sensitive the parameter estimates are to the parametric assumptions and for diagnosing potential misspecification in the imposed network formation model.

4.2 Analysis of linear-in-means peer-effect model

Although Theorem 3 establishes the consistency of the plug-in estimator based on our imputed adjacency matrices, analyzing its convergence rate and inference properties is challenging. First, as discussed in Cai (2022), even when each link contains only i.i.d. classical measurement error, network statistics used in moment conditions typically aggregate errors across links, thereby inducing non-classical measurement error that complicates the analysis. Second, the measurement error generated by our imputation procedure is correlated across links, and because the imputation is nonparametric, its bias and variance may be of the same order, further complicating the analysis of convergence.

Since network links typically enter the moment conditions through network statistics, the asymptotic properties of imputed links established in Theorem 2 could be different from those of the GMM estimators that rely on the imputed network. Therefore, the bandwidth that is optimal for imputing individual links may not be optimal for downstream analysis using network statistics. In the following text, we focus on the linear-in-means peer-effects regression and study the asymptotic behavior of its parameter estimates. We also provide guidance for downstream regression analysis using our imputed networks.

Consider the linear-in-means peer-effects model (4.4) in Example 4, given the weight matrix Σ_M , the GMM estimator is:

$$\hat{\alpha} = \left(\frac{1}{M} \sum_{m=1}^M W'_m Z_m \Sigma_m Z'_m W_m \right)^{-1} \left(\frac{1}{M} \sum_{m=1}^M W'_m Z_m \Sigma_m Z'_m Y_m \right) \quad (4.5)$$

Assumption 6 Suppose that

- (i) **(Exogeneity)** The error vector e_m satisfies $\mathbb{E}(e_m | G_m, W_m) = 0$. Furthermore, we assume that $e_m \perp \epsilon_m | (X_m, \xi_m, W_m)$, where $\epsilon_m := \{\epsilon_{m,ij} | 1 \leq i < j \leq N_m\}$ are error terms in network generating process.
- (ii) **(Asymptotics)** The network sizes $\{N_m | m = 1, \dots, N\}$ are of the same order, and sample sizes $\{n_m | m = 1, \dots, M\}$ are of the same order as well, i.e.,

$$\sup_{M \rightarrow \infty} \inf_{1 \leq m_1, m_2 \leq M} (N_{m_1}/N_{m_2}) > 0, \quad \sup_{M \rightarrow \infty} \inf_{1 \leq m_1, m_2 \leq M} (n_{m_1}/n_{m_2}) > 0$$

- (iii) **(Non-degeneracy)** Let $\lambda_{\min}(\cdot)$ denote the smallest singular value of a matrix, then:

$$\lambda_{\min} \left(\mathbb{E} \left(\frac{1}{N_m} Z'_m W_m \right)' \mathbb{E} \left(\frac{1}{N_m} Z'_m W_m \right) \right) > 0$$

Assumption 6(i) first imposes the strict exogeneity of regressors, $\mathbb{E}(e_m | G_m, W_m) = 0$, which is standard in the literature and rules out correlated effects (Bramoullé, Djebbari and Fortin, 2020). It further assumes that the errors in the linear-in-means peer-effects model, e_m , are conditionally independent of the errors in the network formation process, ϵ_m , which ensures that the imputation error is independent of Y_m .

Assumption 6(ii) requires that all networks have sizes of the same order and that the sample sizes within networks are also of the same order. In other words, no network or sample becomes disproportionately large or disproportionately small.

Assumption 6(iii) imposes a non-degeneracy condition requiring that the smallest singular value of $\mathbb{E} \left(\frac{1}{N_m} Z'_m W_m \right)' \mathbb{E} \left(\frac{1}{N_m} Z'_m W_m \right)$ is bounded away from zero. This requirement is analogous to the relevance condition in instrumental-variable estimation, and is particularly important under large-network

asymptotics. For example, if W_m is i.i.d. within each network and independent of the network (e.g., random treatment assignment), the instrument Z_m becomes asymptotically collinear, causing the smallest singular value of the above matrix to shrink toward zero and leading to an inconsistent IV estimator. In practice, one can let W_m to be correlated with X_m to satisfy this assumption.

Theorem 4 *Under Assumption 6 and conditions in Theorem 2, we have*

$$\sqrt{M}(\hat{\alpha} - \alpha_0 - b_{N,n}(h)) \xrightarrow{d} \mathcal{N}(0, \Omega + v_{N,n}(h))$$

$$\text{where } b_{N,n}(h) = O\left(h^2 + \frac{1}{n^2 h^{2d_\zeta}}\right), \quad v_{N,n}(h) = O\left(\frac{h^4}{n h^{d_\zeta}} + \delta_{N,n}^2 h^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}}\right).$$

The theorem states that $\hat{\alpha} - \alpha_0$ is asymptotically normally distributed, but its limiting distribution is centered away from zero by an $O\left(h^2 + \frac{1}{n^2 h^{2d_\zeta}}\right)$ bias term. The first component, h^2 , arises from the bias of imputed links, while the second term is, in most cases, asymptotically negligible relative to h^2 in most empirically relevant cases⁵. The asymptotic variance can be decomposed into two components: Ω corresponds to the variance that would obtain if the network A were perfectly observed, and $v_{N,n}$ captures the additional variability introduced by the nonparametric estimation of $\hat{A}$. Since the variance component induced by using the imputed network shrinks at rate $1/\sqrt{M}$, whereas the bias does not, this highlights a difference between the asymptotic behavior of the estimator and that of the link-level imputation error.

To illustrate the difference between the asymptotic behavior of the estimator and that of the link-level imputation error, consider the case $d_\zeta \geq 2$, and the optimal bandwidth for minimizing the mean-squared-error of the imputed links, i.e., $h \sim n^{-2/(4+d_\zeta)}$. The bias of $\hat{\alpha}$ is of order $O(h^2)$, which is the same order as the bias of the link imputation error. In contrast, the additional variance induced by the imputed network decreases at the rate $1/M$ relative to the variance of the link imputation error. This implies that, in the downstream regression, the nonparametric estimation bias from the first-step imputation has a much larger impact than the first-step nonparametric estimation variance.

These theoretical observations suggest that reducing the MSE of $\hat{\alpha}$ requires undersmoothing in the imputation step, that is, using a smaller bandwidth h to reduce the bias of the imputed links, rather than attempting to balance the bias and variance of the link-level imputation error. It is worth noting that the existing nonparametric inference literature typically employs undersmoothing for the purpose of constructing valid confidence intervals. In contrast, in our setting, undersmoothing is required even when the objective is to minimize the MSE of $\hat{\alpha}$. This requirement arises from the different asymptotic behaviors of network statistics and link imputation errors.

However, in practice, the asymptotic analysis above has two limitations. First, network sizes are relatively small in most economics applications, so reducing h leaves very few observations available for local imputation. Second, the number of networks M is also small and rarely exceeds 40 in most empirical settings. For these reasons, we recommend that applied researchers start from the optimal bandwidth h selected by cross-validation in the imputation step, then gradually decrease h and examine the corresponding estimates $\hat{\alpha}$. Once the estimates become unstable as h decreases, we should stop undersmoothing at that point.

⁵To see this, note that the optimal bandwidth for minimizing the mean squared error of the imputed links is of order $h^* \sim n^{\frac{-2}{4+d_\zeta}}$. For the second bias component to dominate the first, that is, $h^2 = o\left(\frac{1}{n^2 h^{2d_\zeta}}\right)$, we would need $h = o\left(n^{-\frac{1}{1+d_\zeta}}\right)$, which is much smaller than $(h^*)^2$. Such undersmoothing is rare in practice.

5 Simulation Study

In this section, we examine the finite-sample properties of the proposed method using a series of numerical experiments. We compare its performance with several alternative imputation approaches, including the low-rank imputation methods proposed by Bai and Ng (2021) and Li et al. (2023), and the local PCA method developed by Feng (2023). We also use numerical experiments to illustrate (i) the importance of incorporating observed covariates for improving prediction accuracy, and (ii) the influence of sample splitting on finite-sample performance. We begin with introducing alternative method.

5.1 Alternative Methods

Covariate-only method This method is a dyadic nonparametric regression-based imputation approach that relies solely on the observed covariates X . When unobserved heterogeneity is present, the prediction error of this approach does not diminish as the sample size increases. We compare its numerical performance with that of our proposed method to highlight the importance of accounting for unobserved heterogeneity. We refer to this method as **X** in the following text.

Low-rank imputation When the observed matrix follows a strong-factor structure with an additive noise component and the missing entries exhibit a block-missing pattern⁶ (see (2.3)), Bai and Ng (2021) and Li et al. (2023) propose using low-rank estimation to impute the missing data. The key idea is to first recover the latent factors using the observed submatrices A_{SS} , A_{SS^c} , and A_{S^cS} , and then reconstruct the missing bottom-right submatrix $A_{S^cS^c}$ through the product of the estimated factors.

When the graphon f is nonlinear as in Example 2, the graphon matrix is typically high-rank and thus violates the low-rank assumption required by Bai and Ng (2021) and Li et al. (2023). Although their methods are not theoretically valid in such settings, we include them in our numerical experiments for comparison, given their simplicity and computational efficiency, to assess how our proposed method performs in practice relative to these benchmarks. It is worth noting that both Bai and Ng (2021) and Li et al. (2023) assume that the true number of factors is known a priori. In practice, this information is rarely available. Hence, in our numerical experiments, we select the number of factors via cross-validation.

Local PCA The local PCA method proposed by Feng (2023) is suitable for high-rank graphon. Specifically, it first finds the neighbors of each node i based on the pseudo-distance using k -nearest neighbors (kNN), and then performs low-rank imputation on the submatrix formed by node i 's neighbors to impute the missing entries. As discussed earlier, when the underlying graphon is smoother than twice continuously differentiable, the local PCA method can, in theory, achieve a higher-order approximation error. We compare our method with local PCA to assess whether this theoretical advantage persists in finite samples.

A practical concern is that the local PCA method requires prior knowledge of the latent dimension d_ζ , since this determines the rank when performing PCA on each submatrix. However, d_ζ is typically unknown and rarely supported by prior knowledge. While Feng (2023) propose an ad hoc procedure in their numerical simulations to address this issue, we instead use cross-validation to jointly select both the number of neighbors k in the kNN step and the rank used in PCA.

It is worth noting that local PCA does not incorporate observed covariates⁷. In our numerical experiments, we therefore consider two implementations of their method: (1) **LPCA**, which ignores

⁶Bai and Ng (2021) refer to this pattern as a tall-wide structure in the context of panel data, while Li et al. (2023) study an analogous setting in network data and refer to it as egocentric sampling.

⁷Feng (2023) briefly discuss how to include covariates in a linear form in their appendix.

covariates X and applies local PCA directly to the adjacency matrix, and (2) **X-LPCA**, which first removes the variation explained by X and then applies local PCA to the residual matrix.

Local TWFE without covariates We also consider an alternative method that directly applies our local TWFE approach to the adjacency matrix without extracting the variation explained by X . As discussed earlier, because the information about ζ recovered from observed data is noisy, it is unwise to ignore the observed covariates and absorb them into the latent factors. We include the numerical simulation results of this method to evaluate the importance of incorporating covariate information. We refer to this method as **LTWFE** and to our proposed method as **X-LTWFE** in the following text.

Sample splitting While our asymptotic analysis relies on sample splitting for theoretical tractability, this requirement is primarily technical. In practice, sample splitting reduces the effective sample size, and since network sizes are relatively small in most economics applications, its finite-sample performance can be worse than that of the full-sample implementation. Accordingly, we implement all alternative methods without sample splitting.

For our proposed method, we consider both versions, with and without sample splitting, to examine how this choice affects finite-sample performance. We refer to our method with sample splitting as **X-LTWFE-SP** in the following text.

Summary We summarize all imputation methods used in the numerical experiments below:

- (i) **(X)** Covariate-only nonparametric method that imputes missing links using X alone.
- (ii) **(LR)** The low-rank imputation methods of [Bai and Ng \(2021\)](#) and [Li et al. \(2023\)](#).
- (iii) **(LPCA)** The local PCA method that ignores covariates.
- (iv) **(X-LPCA)** First removes the variation explained by X and then applies local PCA to the residual matrix.
- (v) **(LTWFE)** The local two-way fixed-effects method that ignores covariates.
- (vi) **(X-LTWFE)** Our proposed method. It first removes the variation explained by X and then applies the local two-way fixed-effects regression to the residual matrix.
- (vii) **(X-LTWFE-SP)** Our proposed method with sample-splitting.

5.2 Simulation for Imputation Accuracy

The network in the simulation is generated as

$$A_{ij} = \mathbf{1}(\omega(X_i, X_j)' \beta + g(\xi_i, \xi_j) - U_{ij} \geq 0) \quad (5.1)$$

$$\omega(X_i, X_j)' = ((X_{i1} - X_{j1})^2, (X_{i2} - X_{j2})^2), \quad g(\xi_i, \xi_j) = \xi_{i1} + \xi_{j1} - \frac{1}{8}(\xi_{i2} - \xi_{j2})^2$$

The dimensions of the observable and latent characteristics are $d_X = d_\xi = 2$, and $\{(X_i, \xi_i) \mid i = 1, \dots, N\}$ is i.i.d. across individuals. Covariates and latent factors are generated according to

$$X_{id} = \frac{1}{2}(\xi_{i1} + \xi_{i2}) + \epsilon_{X,id}, \quad \forall d = 1, 2$$

and

$$\xi_i \sim \mathcal{N}\left(0, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right), \quad \epsilon_{X,i1}, \epsilon_{X,i2} \sim \text{Uniform}([-1, 1]), \quad \epsilon_{X,i1} \perp \epsilon_{X,i2}$$

The idiosyncratic shocks $\{U_{ij}\}_{1 \leq i < j \leq N}$ are independently drawn from a type-I extreme-value distribution. The coefficient vector β captures homophily effects on observables and takes values $\beta \in \{(-0.5, 0.5), (-2, -2)\}$. We vary the magnitude of β to control the sparsity of the network: for example, when $\beta = (-0.5, 0.5)$, the resulting network tends to be denser, with a larger number of links being formed.

The network size is $N = 200$, and the simulations consider sampling rates $\varphi := n/N \in \{0.2, 0.3, 0.4, 0.5, 0.6, 0.8\}$ to compare the performance of the proposed and alternative methods under different sampling scenarios. We conduct $S = 1,000$ simulations in the numerical experiment, and for each simulation s we construct the imputed network $\hat{A}_s$ and compute the mean squared error (MSE) between $\hat{A}_s$ and the underlying graphon f_s on the missing subnetwork, namely $\frac{1}{(N-n)(N-n-1)} \sum_{i,j \in S^c} (\hat{A}_{s,ij} - f_{s,ij})^2$. The reported statistic is the root mean-squared error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{S} \sum_{s=1}^S \frac{1}{(N-n)(N-n-1)} \sum_{i,j \in S^c} (\hat{A}_{s,ij} - f_{s,ij})^2} \quad (5.2)$$

Table 1: Simulation results

	X	LR	LPCA	LTWFE	X-LPCA	X-LTWFE	X-LTWFE-E-SP
	($\times 0.01$)	($\times 0.01$)	($\times 0.01$)	($\times 0.01$)	($\times 0.01$)	($\times 0.01$)	($\times 0.01$)
$\beta = (-0.5, -0.5)$							
$\varphi = 0.2$	21.0	15.8	15.9	15.3	14.9	10.9	13.1
$\varphi = 0.3$	20.9	13.9	14.0	13.9	12.3	9.7	11.3
$\varphi = 0.4$	20.8	12.4	12.5	13.0	10.8	8.9	10.4
$\varphi = 0.5$	20.8	11.3	11.3	12.3	9.8	8.4	9.7
$\varphi = 0.6$	20.8	10.3	10.4	11.9	9.2	8.1	9.3
$\varphi = 0.8$	20.6	9.2	9.2	11.0	8.3	7.5	8.6
$\beta = (-2, -2)$							
$\varphi = 0.2$	13.8	16.3	16.4	15.3	12.7	10.2	11.3
$\varphi = 0.3$	13.7	14.8	14.9	14.0	11.5	9.4	10.3
$\varphi = 0.4$	13.7	13.3	13.5	13.1	10.5	8.9	9.7
$\varphi = 0.5$	13.6	11.7	11.8	12.5	9.8	8.6	9.3
$\varphi = 0.6$	13.6	10.5	10.6	11.9	9.3	8.3	9.0
$\varphi = 0.8$	13.5	9.2	9.2	11.0	8.4	7.8	8.5

Note: We conduct Monte Carlo simulations based on 1,000 repetitions of the network formation model (5.1). The network size is $N = 200$. We set $\beta \in \{(-0.5, 0.5), (-2, -2)\}$ and sampling rates $\varphi \in \{0.2, 0.3, 0.4, 0.5, 0.6, 0.8\}$. We report RMSE as in (5.2) for the following imputation methods: (1) covariate-only nonparametric imputation that imputes missing outcomes using X alone (**X**), (2) the low-rank imputation methods of Bai and Ng (2021) and Li et al. (2023) (**LR**), (3) the local PCA method that ignores covariates (**LPCA**), (4) the local PCA method using covariates (**X-LPCA**), (5) the local two-way fixed-effects regression that ignores covariates (**LTWFE**), (6) the local two-way fixed-effects regression using covariates (**X-LTWFE**), and (7) X-LTWFE method with sample-splitting (**X-LTWFE-SP**). It is important to note that LR, LPCA, LTWFE, X-LPCA, and X-LTWFE are implemented without sample splitting. RMSEs are presented in units of 0.01 for ease of reading.

We evaluate the performance of our imputation method and alternative approaches in both dense networks (e.g., $\beta = (-0.5, -0.5)$, for which the average probability of forming a link is about 34%) and sparse networks (e.g., $\beta = (-2, -2)$, for which the average probability of forming a link is about 10%). The simulation results are reported in Table 1, with RMSEs presented in units of 0.01 for ease of reading.

The first column (X) reports the RMSE of covariate-only method, which imputes missing links using X alone. Because this method ignores unobserved heterogeneity, its imputations are systematically biased, and the error does not vanish as the network size or sampling rate increases. Consistent with the theoretical prediction, the numerical results show that, as the sampling rate φ increases, the RMSE of the covariate-only method exhibits very limited improvement (from 0.210 to 0.206 in the dense scenario, and from 0.138 to 0.135 in the sparse scenario).

The second column (LR) reports the RMSE of the low-rank imputation, and the third column (LPCA) reports the RMSE of the local PCA. Since the underlying graphon in (5.1) is high-rank, local PCA should, in theory, outperform the low-rank approach. However, the numerical results indicate that the RMSEs of the two methods are nearly the same across all scenarios. We conjecture that this pattern is driven by two reasons: (1) Although the graphon is not exactly low-rank, it is approximately low-rank, so the low-rank approach still provides a good approximation to the graphon matrix, especially when performance is measured using Frobenius norm. (2) Local PCA relies only on the subnetwork formed by neighbors, which leads to substantial sample loss and limits its performance in finite samples. In line with our theory, the local PCA estimator outperforms the covariate-only (X) method in most cases, and its prediction error decreases as the sampling rate increases.

The fourth column (LTWFE) reports the RMSE of the local two-way fixed effects imputation that ignores covariates. In line with the theory, its prediction error decreases as n increases. Table 1 shows that at low sampling rates, the LTWFE method performs better than local PCA, whereas once the sampling rate becomes large (e.g., when φ exceeds 40%), local PCA outperforms LTWFE. This suggests that although local PCA has better asymptotic properties, in the sense that it achieves higher-order approximation error, it requires a larger sample size to realize this advantage, whereas LTWFE is more robust in small samples. In addition, the LTWFE method is more robust to network sparsity: as shown in Table 1, its performance is nearly unaffected by changes in the sparsity of the adjacency matrix.

The fifth column (X-LPCA) and the sixth column (X-LTWFE) report the performance of LPCA and LTWFE when incorporating covariate information. These methods first extract the variation of the missing links that can be explained by X , and then apply LPCA and LTWFE to the residual matrix, respectively. Compared with LPCA and LTWFE, incorporating covariate information leads to substantial improvements in prediction accuracy, particularly for X-LTWFE. The gains are especially large when the sampling rate is low or when the network is sparse. X-LPCA exhibits a similar pattern. In addition, Table 1 shows that the proposed method outperforms all alternatives in our simulations, and its performance remains robust even under sparse networks and low sampling rates, which are common in empirical applications.

The last column (X-LTWFE-SP) reports the performance of our proposed estimator with sample splitting. As discussed earlier, although sample-splitting facilitates asymptotic analysis, it reduces the effective sample size, and this issue can be severe in small-sample settings. In addition, it is reasonable to expect (though we do not formally prove it) that the correlation between the pseudo-distance and the error terms diminishes, making sample splitting unnecessary in theory. Table 1 shows that performance of proposed imputation under sample splitting is always worse than the performance without splitting the sample, but the difference between the two methods becomes smaller as the sample size increases.

In summary, the simulations show that our proposed method outperforms the alternative methods and is more robust across different levels of network sparsity and sampling rates. In addition, for finite samples, we recommend using the version without sample splitting in order to achieve higher prediction

accuracy. Further numerical results are provided in the Appendix.

5.3 Simulation for estimates of parameter

In this subsection, we evaluate the performance of parameter estimates in downstream analysis using the imputed network. We compare the performance of our method with the alternative methods introduced previously. We focus on two of the most commonly used types of economic analysis based on network data:

Regression on network statistics Suppose for each network $m = 1, \dots, M$, the outcome Y_{mi} is generated according to:

$$Y_{mi} = \alpha_0 + \alpha_1 \phi_{mi}(A_m) + e_{mi} \quad (5.3)$$

where $\phi_{mi} : [0, 1]^{N_m \times N_m} \rightarrow \mathbb{R}$ computes the degree centrality for node i . For example, for any matrix $\tilde{A} \in [0, 1]^{N_m \times N_m}$, if $\phi_{mi}(\tilde{A}) = \frac{1}{N_m} \sum_{j=1}^{N_m} \tilde{A}_{ij}$, then $\phi_{mi}(\cdot)$ represents the normalized degree of individual i when $\tilde{A}$ is an adjacency matrix. Another example is eigenvector centrality. For any matrix $\tilde{A} \in [0, 1]^{N_m \times N_m}$, consider the eigenvalue decomposition: A

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6 Empirical Application

7 Conclusion

Sampled network data are common in empirical research because collecting full network information is costly, but using sampled networks can lead to biased estimates. We propose a flexible imputation method for sampled networks and show that downstream empirical analysis using the imputed network yields consistent parameter estimates.

Our method separately imputes the part of each missing link that is explained by covariates and the remaining variation not captured by them: the former via projection onto observed covariates, and the latter using local two-way fixed effects. The imputation method avoids parametric assumptions, does not rely on low-rank restrictions, and flexibly incorporates observed covariates and unobserved heterogeneity. We establish entrywise convergence rates for the imputed matrix and prove consistency of GMM estimators based on the imputed network. We further derive the convergence rate of the corresponding estimator in the linear-in-mean peer effects model. Simulations show strong performance of our method both in terms of imputation accuracy and in downstream empirical analysis. We illustrate our method with an application to the microfinance dataset of [Banerjee et al. \(2013\)](#).

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A Implementation

A.1 1

Algorithm 2 Imputation with Cross Validation

Input: Kernel $K(\cdot)$, a list of bandwidths $\{h_b\}_{b=1,\dots,B}$, incomplete network A^{obs} , covariate matrix X .

Output: Imputed network $\hat{A}$.

Step 1: Extract information explained by X

Run dyadic nonparametric regression to obtain $\hat{\Pi}_X(\cdot, \cdot)$. Also calculate $\hat{\Pi}_{X,i'j'}^\perp = A_{i'j'} - \hat{\Pi}_{X,i'j'}$ for each observed entry (i', j') , i.e., $i' \in \mathcal{S} \vee j' \in \mathcal{S}$.

Step 2: Calculate similarity

For each $i \in 1, \dots, N$ and $i' \in \mathcal{S}$, calculate the pseudo distance using observed adjacency matrix:

$$\hat{d}_{ii'} = \max_{k=1,\dots,n} \left| \frac{1}{n} \sum_{\ell=1}^n A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|$$

Step 3: Leave-one-out cross validation

For each $h_b \in \{h_1, \dots, h_B\}$, and for each $i \in \mathcal{S}$ and $j \in \mathcal{S}^c$, run the local (weighted) two-way fixed effect regression

$$\hat{\alpha}(h_b) \in \underset{\alpha \in \mathbb{R}^{n+1}}{\text{argmin}} \sum_{i' \in \mathcal{S}, j' \in \mathcal{S} \cup \{j\} \setminus \{i\}} \left(\hat{\Pi}_{X,i'j'}^\perp - \alpha_{i'} - \alpha_{j'} \right)^2 K_{h_b}(\hat{d}_{ii'}) K_{h_b}(\hat{d}_{jj'})$$

to obtain $\tilde{A}_{ij}(h_b) = \hat{\alpha}_i(h_b) + \hat{\alpha}_j(h_b) + \hat{\Pi}_{X,ij}$. Truncate $\tilde{A}_{ij}(h_b)$ to get

$$\hat{A}_{ij}(h_b) = \begin{cases} \tilde{A}_{ij}(h_b), & \text{if } \tilde{A}_{ij}(h_b) \in [0, 1] \\ 1 & \text{if } \tilde{A}_{ij}(h_b) > 1 \\ 0 & \text{if } \tilde{A}_{ij}(h_b) < 0 \end{cases}$$

Then the optimal h^* solves

$$h^* = \underset{h_b \in \{h_1, \dots, h_B\}}{\text{argmin}} \sum_{i \in \mathcal{S}, j \in \mathcal{S}^c} \left(A_{ij} - \hat{A}_{ij}(h_b) \right)^2$$

Step 4: Local imputation

For each $i \in \mathcal{S}^c$ and $j \in \mathcal{S}^c$, run the local (weighted) two-way fixed effect regression

$$\hat{\alpha} \in \underset{\alpha \in \mathbb{R}^{n+2}}{\text{argmin}} \sum_{i' \in \mathcal{S} \cup \{i\}, j' \in \mathcal{S} \cup \{j\}} \left(\hat{\Pi}_{X,i'j'}^\perp - \alpha_{i'} - \alpha_{j'} \right)^2 K_{h^*}(\hat{d}_{ii'}) K_{h^*}(\hat{d}_{jj'})$$

to obtain $\tilde{A}_{ij} = \hat{\alpha}_i + \hat{\alpha}_j + \hat{\Pi}_{X,ij}$. Truncate $\tilde{A}_{ij}$ to get

$$\hat{A}_{ij} = \begin{cases} \tilde{A}_{ij}, & \text{if } \tilde{A}_{ij} \in [0, 1] \\ 1 & \text{if } \tilde{A}_{ij} > 1 \\ 0 & \text{if } \tilde{A}_{ij} < 0 \end{cases}$$

and zero the diagonal of $\hat{A}$ such that $\hat{A}_{ii} = 0$ for all i .

Algorithm 3 Imputation with Cross Validation and Sample Splitting

Input: Kernel $K(\cdot)$, a list of bandwidths $\{h_b\}_{b=1,\dots,B}$, incomplete network A^{obs} , covariate matrix X .

Random split $\mathcal{S}$ into two parts, $\mathcal{S}_1, \mathcal{S}_2$, with equal sizes.

Output: Imputed network $\hat{A}$.

Step 1: Extract information explained by X

Run dyadic nonparametric regression to obtain $\hat{\Pi}_X(\cdot, \cdot)$. Also calculate $\hat{\Pi}_{X,i'j'}^\perp = A_{i'j'} - \hat{\Pi}_{X,i'j'}$ for each observed entry (i', j') , i.e., $i' \in \mathcal{S} \vee j' \in \mathcal{S}$.

Step 2: Calculate similarity

Calculate the pseudo distance using observed adjacency matrix:

$$\hat{d}_{ii'} = \max_{k \in \mathcal{S}_1} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|, \quad i \in \mathcal{S}_2 \cup \mathcal{S}^c, i' \in \mathcal{S}_2$$

Step 3: Leave-one-out cross validation

For each $h_b \in \{h_1, \dots, h_B\}$, and for each $i \in \mathcal{S}_2$ and $j \in \mathcal{S}^c$, run the local (weighted) two-way fixed effect regression

$$\hat{\alpha}(h_b) \in \underset{\alpha \in \mathbb{R}^{|\mathcal{S}_2|+1}}{\text{argmin}} \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2 \cup \{j\} \setminus \{i\}} \left(\hat{\Pi}_{X,i'j'}^\perp - \alpha_{i'} - \alpha_{j'} \right)^2 K_{h_b}(\hat{d}_{ii'}) K_{h_b}(\hat{d}_{jj'})$$

to obtain $\tilde{A}_{ij}(h_b) = \hat{\alpha}_i(h_b) + \hat{\alpha}_j(h_b) + \hat{\Pi}_{X,ij}$. Truncate $\tilde{A}_{ij}(h_b)$ to get

$$\hat{A}_{ij}(h_b) = \begin{cases} \tilde{A}_{ij}(h_b), & \text{if } \tilde{A}_{ij}(h_b) \in [0, 1] \\ 1 & \text{if } \tilde{A}_{ij}(h_b) > 1 \\ 0 & \text{if } \tilde{A}_{ij}(h_b) < 0 \end{cases}$$

Then the optimal h^* solves

$$h^* = \underset{h_b \in \{h_1, \dots, h_B\}}{\text{argmin}} \sum_{i \in \mathcal{S}_2, j \in \mathcal{S}^c} \left(A_{ij} - \hat{A}_{ij}(h_b) \right)^2$$

Step 4: Local imputation

For each $i \in \mathcal{S}^c$ and $j \in \mathcal{S}^c$, run the local (weighted) two-way fixed effect regression

$$\hat{\alpha} \in \underset{\alpha \in \mathbb{R}^{|\mathcal{S}_2|+2}}{\text{argmin}} \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \left(\hat{\Pi}_{X,i'j'}^\perp - \alpha_{i'} - \alpha_{j'} \right)^2 K_{h^*}(\hat{d}_{ii'}) K_{h^*}(\hat{d}_{jj'})$$

to obtain $\tilde{A}_{ij} = \hat{\alpha}_i + \hat{\alpha}_j + \hat{\Pi}_{X,ij}$. Truncate $\tilde{A}_{ij}$ to get

$$\hat{A}_{ij} = \begin{cases} \tilde{A}_{ij}, & \text{if } \tilde{A}_{ij} \in [0, 1] \\ 1 & \text{if } \tilde{A}_{ij} > 1 \\ 0 & \text{if } \tilde{A}_{ij} < 0 \end{cases}$$

and zero the diagonal of $\hat{A}$ such that $\hat{A}_{ii} = 0$ for all i .

B Consistency of Imputation

$$\sup_f \|f\|_\infty \leq \rho_f$$

Proof of Lemma 1. Recall that

$$\hat{d}_{ii'} = \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right|$$

and its population counterpart:

$$d_{ii'} = \sup_{\zeta_k} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell) (f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right|$$

and define

$$\tilde{d}_{ii'} = \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} f(\zeta_k, \zeta_\ell) (f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) \right|$$

then by triangular inequality:

$$\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\hat{d}_{ii'} - d_{ii'}| \leq \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\hat{d}_{ii'} - \tilde{d}_{ii'}| + \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\tilde{d}_{ii'} - d_{ii'}|$$

Bound on $\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\hat{d}_{ii'} - \tilde{d}_{ii'}|$ Note that for i, i', k , we have:

$$\begin{aligned} & \left| \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} A_{k\ell} (A_{i\ell} - A_{i'\ell}) \right| - \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} f(\zeta_k, \zeta_\ell) (f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) \right| \right| \\ & \leq \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell} (f(\zeta_i, \zeta_\ell) + f(\zeta_{i'}, \zeta_\ell)) \right| + \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} f(\zeta_k, \zeta_\ell) (\epsilon_{i\ell} - \epsilon_{i'\ell}) \right| \\ & \quad + \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell} (\epsilon_{i\ell} - \epsilon_{i'\ell}) \right| + \frac{2\rho_f^2}{|\mathcal{S}_1|} \end{aligned}$$

which implies

$$\begin{aligned} \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\hat{d}_{ij} - \tilde{d}_{ij}| & \leq \underbrace{\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell} (f(\zeta_i, \zeta_\ell) + f(\zeta_{i'}, \zeta_\ell)) \right|}_{B_1} \\ & \quad + \underbrace{\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} f(\zeta_k, \zeta_\ell) (\epsilon_{i\ell} - \epsilon_{i'\ell}) \right|}_{B_2} \\ & \quad + \underbrace{\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell} (\epsilon_{i\ell} - \epsilon_{i'\ell}) \right|}_{B_3} + \frac{2\rho_f^2}{|\mathcal{S}_1|} \end{aligned}$$

We employ Bernstein's concentration bound, the conditional independence of $\{\epsilon_{ij} \mid i, j = 1, \dots, N\}$, and $|\mathcal{S}_1| = |\mathcal{S}_2| = n/2$ to show that for any $t > 0$

$$\begin{aligned}\mathbb{P}(B_1 \geq t) &\leq N^3 \mathbb{P}\left(\left|\frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell}(f(\zeta_i, \zeta_\ell) + f(\zeta_{i'}, \zeta_\ell))\right| \geq t\right) \leq 2N^3 \exp\left(-\frac{nt^2}{16\sigma^2\rho_f^2 + 8\sigma\rho_f t}\right) \\ \mathbb{P}(B_2 \geq t) &\leq N^3 \mathbb{P}\left(\left|\frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} f(\zeta_k, \zeta_\ell)(\epsilon_{i\ell} - \epsilon_{i'\ell})\right| \geq t\right) \leq 2N^3 \exp\left(-\frac{nt^2}{16\sigma^2\rho_f^2 + 8\sigma\rho_f t}\right) \\ \mathbb{P}(B_3 \geq t) &\leq N^3 \mathbb{P}\left(\left|\frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1 \setminus \{k\}} \epsilon_{k\ell}(\epsilon_{i\ell} - \epsilon_{i'\ell})\right| \geq t\right) \leq 2N^3 \exp\left(-\frac{nt^2}{16\sigma^4 + 8\sigma^2 t}\right)\end{aligned}$$

Then we can find a sufficiently large constant $C_1 > 0$ such that when $t = C_1 \frac{\log N}{\sqrt{n}}$, we have

$$\mathbb{P}\left(\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\hat{d}_{ii'} - \tilde{d}_{ii'}| \geq C_1 \frac{\log N}{\sqrt{n}} + \frac{4\rho_f^2}{n}\right) \leq \frac{1}{3} n^{-\frac{1}{2}}$$

Bound on $\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\tilde{d}_{ii'} - d_{ii'}|$ Note that for i, i', k , we have:

$$\begin{aligned}|\tilde{d}_{ij} - d_{ij}| &\leq \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) - \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| \\ &\quad + \sup_{\zeta_k} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| \\ &\quad - \max_{k \notin \{i, i'\}} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right|\end{aligned}$$

and $\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} |\tilde{d}_{ii'} - d_{ii'}|$ is bounded by:

$$\begin{aligned}&\underbrace{\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \max_{k \notin \{i, i'\}} \left| \frac{1}{|\mathcal{S}_1|} \sum_{\ell \in \mathcal{S}_1} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) - \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right|}_{Q_1} \\ &+ \underbrace{\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \left(\sup_{\zeta_k} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| - \max_{k \notin \{i, i'\}} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell)(f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| \right)}_{Q_2}\end{aligned}$$

One can employ Bernstein's inequality (Lemma D.3) to show that for any $t > 0$:

$$\mathbb{P}(Q_1 \geq t) \leq 2N^3 \exp\left(-\frac{nt^2}{16\rho_f^4 + 8\sigma\rho_f^4 t}\right)$$

Then we can find a sufficiently large constant $C_2 > 0$ such that when $t = C_2 \frac{\log N}{\sqrt{n}}$, we have

$$\mathbb{P}\left(Q_1 \geq C_2 \frac{\log N}{\sqrt{n}} + \frac{4\rho_f^2}{n}\right) \leq \frac{1}{3} n^{-\frac{1}{2}}$$

For Q_2 , suppose ζ^* the maximizer of $d_{ii'}$, we have

$$\begin{aligned}
S_2 &\leq \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \left(\left| \int_{\zeta_\ell} f(\zeta^*, \zeta_\ell) (f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| - \min_{k \notin \{i, i'\}} \left| \int_{\zeta_\ell} f(\zeta_k, \zeta_\ell) (f(\zeta_i, \zeta_\ell) - f(\zeta_{i'}, \zeta_\ell)) d\mathbb{P}_\xi(\zeta_\ell) \right| \right) \\
&\stackrel{(i)}{\leq} \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \min_{k \notin \{i, i'\}} 2\rho_f L d_\zeta^{1/2} \|\zeta^* - \zeta_k\| \\
&\leq 2\rho_f L d_\zeta^{1/2} \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \sup_{\zeta} \min_{k \notin \{i, i'\}} \|\zeta - \zeta_k\|
\end{aligned}$$

where inequality (i) follows from the Assumption 1(v) that since f is Lipschitz function with a constant $L > 0$. By Assumption 1(iv), there exists a t -covering of the support of ζ that consists of $\mathcal{J} \leq 4^{d_\zeta} d_\zeta^{d_\zeta/2} t^{-d_\zeta}$ open balls with radius t , denoted as $\{\bar{\zeta}_j\}_{j=1}^{\mathcal{J}}$. Therefore, we immediately have:

$$\sup_{\zeta} \min_{k \notin \{i, i'\}} \|\zeta - \zeta_k\| \leq \max_{j=1, \dots, \mathcal{J}} \min_{k \notin \{i, i'\}} \|\bar{\zeta}_j - \zeta_k\| + t$$

It follows that:

$$\begin{aligned}
\mathbb{P} \left(\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \sup_{\zeta} \min_{k \notin \{i, i'\}} \|\zeta - \zeta_k\| \geq 2t \right) &\leq N^2 \mathbb{P} \left(\max_{j=1, \dots, \mathcal{J}} \min_{k \notin \{i, i'\}} \|\bar{\zeta}_j - \zeta_k\| \geq t \right) \\
&\leq N^2 \mathcal{J} \mathbb{P} \left(\min_{k \notin \{i, i'\}} \|\bar{\zeta}_j - \zeta_k\| \geq t \right) \\
&\stackrel{(i)}{=} N^2 \mathcal{J} \left(\mathbb{P} \left(\min_{k \notin \{i, i'\}} \|\bar{\zeta}_j - \zeta_k\| \geq t \right) \right)^N \\
&\stackrel{(ii)}{\leq} N^2 \mathcal{J} (1 - \underline{c} t^{d_\zeta})^N \\
&\leq 4^{d_\zeta} d_\zeta^{d_\zeta/2} N^2 t^{-d_\zeta} (1 - \underline{c} t^{d_\zeta})^N
\end{aligned}$$

where equality (i) comes from the independence of $\{\zeta_i \mid i = 1, \dots, N\}$, and inequality (ii) follows from Assumption 1(iv). Then we can find a constant $C_3 > 0$ such that when $t = C_3 \frac{\log N}{N^{1/d_\zeta}}$,

$$\mathbb{P} \left(Q_2 \geq C_3 \frac{\log N}{N^{1/d_\zeta}} \right) \leq \frac{1}{3} n^{-\frac{1}{2}}$$

Combine all these above, and let $\gamma_1 = C_3$ as $\gamma_2 = C_1 + C_2$, then

$$\mathbb{P} \left(\max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \left| \hat{d}_{ij} - d_{ij} \right| \geq \gamma_1 \frac{\log N}{N^{1/d_\zeta}} + \gamma_2 \frac{\log N}{\sqrt{n}} + \frac{6\rho_f^2}{N_0} \right) \leq n^{-\frac{1}{2}}$$

and we complete the proof. It is sufficiently focusing on the event

$$\mathcal{A} := \left\{ \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \left| \hat{d}_{ii'} - d_{ii'} \right| \geq \gamma_1 \frac{\log N}{N^{1/d_\zeta}} + \gamma_2 \frac{\log N}{\sqrt{n}} \right\} = \left\{ \max_{i \in \mathcal{S}^c, i' \in \mathcal{S}_2} \left| \hat{d}_{ii'} - d_{ii'} \right| \geq \delta_{N,n} \right\} \quad (\text{B.1})$$

■

Lemma B.1 Under event $\mathcal{A}$ as in (B.1), for any $h > 0$ such that $\delta_{N,n}/h \rightarrow 0$, then for all $i \in \mathcal{S}^c, i' \in \mathcal{S}_2$, when $\hat{d}_{ii'} \leq h$, there is a constant $\eta > 0$ that

$$\|\zeta_i - \zeta_j\| \leq \eta h$$

Proof of Lemma B.1. Note that for any $i \in \mathcal{S}^c, i' \in \mathcal{S}_2$, since $\delta_{N,n}/h \rightarrow 0$, there exists constant

$C_1 > 1$ such that

$$\{\hat{d}_{ii'} \leq h\} \subseteq \{d_{ii'} \leq h + \delta_{N,n}\} \subseteq \{d_{ii'} \leq C_1 h\}$$

Also, by Assumption 2, we have

$$\{d_{ii'} \leq C_1 h\} \subseteq \{\|\zeta_i - \zeta_{i'}\| \leq \lambda^{-1} C_1 h\}$$

Let $\eta = \lambda^{-1} C_1$, and we complete the proof. ■

Proof of Theorem 2. In the proof we focus on the event $\mathcal{A}$ as in (B.1). Since $\mathbb{P}(\mathcal{A}^c) \leq n^{-1/2}$ and both the bias and variance on $\mathcal{A}^c$ are of strictly smaller asymptotic order than those in Theorem 2, their contribution is negligible.

Bias Fixed $i, j \in \mathcal{S}^c$, for any $i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}$, for notational simplicity, define

$$\begin{aligned} K_{h,i'} &:= K(d_{ii'}/h), \quad K_{h,j'} := K(d_{jj'}/h), \quad K_{h,i'j'} := K_{h,i'} K_{h,j'} \\ \hat{K}_{h,i'} &:= K(\hat{d}_{ii'}/h), \quad \hat{K}_{h,j'} := K(\hat{d}_{jj'}/h), \quad \hat{K}_{h,i'j'} := \hat{K}_{h,i'} \hat{K}_{h,j'} \end{aligned}$$

The two-way fixed-effect estimator using weights is:

$$\begin{aligned} \hat{A}_{ij} &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'j} A_{i'j}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'j}} + \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij'} A_{ij'}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij'}} - \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} A_{i'j'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'}} \\ &= \underbrace{\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} f_{i'j}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} + \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'} f_{ij'}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'}} - \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} f_{i'j'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'}}}_{Q_1} \\ &\quad + \underbrace{\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \epsilon_{i'j}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} + \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'} \epsilon_{ij'}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'}} - \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} \epsilon_{i'j'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'}}}_{Q_2} \end{aligned}$$

By Assumption 1(v), the Taylor expansion of $f_{i'j'}$ reads:

$$\begin{aligned} f_{i'j'} &= f(\zeta_i, \zeta_j) + \nabla'_{\zeta_i} f(\zeta_i, \zeta_j)(\zeta_{i'} - \zeta_i) + \nabla'_{\zeta_j} f(\zeta_i, \zeta_j)(\zeta_{j'} - \zeta_j) \\ &\quad + (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' \nabla^2 f(\zeta_i + \eta'_{1,i'j'}(\zeta_{i'} - \zeta_i), \zeta_j + \eta'_{2,i'j'}(\zeta_{j'} - \zeta_j))(\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j) \end{aligned}$$

where $\eta_{1,i'j'}, \eta_{2,i'j'} \in [0, 1]^{d_\zeta}$, then it is straightforward to verify that ⁸

$$\begin{aligned} Q_1 &= f(\zeta_i, \zeta_j) + \nabla'_{\zeta_i} f(\zeta_i, \zeta_j)(\bar{\zeta}_i - \zeta_i) + \hat{R}_1 \\ &\quad + f(\zeta_i, \zeta_j) + \nabla'_{\zeta_j} f(\zeta_i, \zeta_j)(\bar{\zeta}_j - \zeta_j) + \hat{R}_2 \\ &\quad - f(\zeta_i, \zeta_j) - \nabla'_{\zeta_i} f(\zeta_i, \zeta_j)(\bar{\zeta}_i - \zeta_i) - \nabla'_{\zeta_j} f(\zeta_i, \zeta_j)(\bar{\zeta}_j - \zeta_j) + \hat{R}_3 \\ &= f(\zeta_i, \zeta_j) + (\hat{R}_1 + \hat{R}_2 + \hat{R}_3) \end{aligned}$$

⁸Define

$$\bar{\zeta}_i = \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \zeta_{i'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}}, \quad \bar{\zeta}_j = \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'} \zeta_{j'}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'}}$$

The remaining quadratic terms $\hat{R}_1, \hat{R}_2, \hat{R}_3$ are:

$$\begin{aligned}\hat{R}_1 &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}}, \quad \hat{R}_2 = \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'} (\zeta_j - \zeta_{j'})' H_{j'} (\zeta_j - \zeta_{j'})}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'}}, \\ \hat{R}_3 &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'}}\end{aligned}$$

where

$$\begin{aligned}H_{i'} &= \nabla_{\zeta_i, \zeta_i}^2 f(\zeta_i + \eta'_{1,i'j} (\zeta_{i'} - \zeta_i), \zeta_j), \quad H_{j'} = \nabla_{\zeta_j, \zeta_j}^2 f(\zeta_i, \zeta_j + \eta'_{2,ij'} (\zeta_{j'} - \zeta_j)) \\ H_{i'j'} &= \nabla^2 f(\zeta_i + \eta'_{1,i'j} (\zeta_{i'} - \zeta_i), \zeta_j + \eta'_{2,ij'} (\zeta_{j'} - \zeta_j))\end{aligned}$$

Under boundedness condition of f 's second-order derivative (Assumption 1(v)), Lemma B.1, and the compact support assumption (Assumption 4), there exists $0 \leq \rho_{f''} < \infty$, such that

$$\hat{R}_1 = \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \mathbf{1}(\|\zeta_i - \zeta_{i'}\| \leq \eta h) (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} \leq \rho_{f''} \eta^2 h^2$$

By the same argument we can also show that $\hat{R}_2 \leq \rho_{f''} \eta^2 h^2$ and $\hat{R}_3 \leq 4\rho_{f''} \eta^2 h^2$. Thus,

$$\mathbb{E}(Q_1 - f_{ij} \mid \zeta) \leq 6\rho_{f''} \eta^2 h^2$$

In addition, sample-splitting guarantees that $\mathbb{E}(Q_2 \mid \zeta) = 0$, and therefore,

$$\mathbb{E}(\hat{A}_{ij} - f_{ij} \mid \zeta) \leq 6\rho_{f''} \eta^2 h^2$$

Variance Define

$$\begin{aligned}R_1 &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'}}, \quad R_2 = \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j'} (\zeta_j - \zeta_{j'})' H_{j'} (\zeta_j - \zeta_{j'})}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j'}}, \\ R_3 &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,i'j'}}\end{aligned}$$

and

$$\begin{aligned}&\text{Var}(\hat{A}_{ij} - \mathbb{E}(\hat{A}_{ij} \mid \zeta)) \\ &\leq 3\text{Var}(R_1 + R_2 + R_3) + 3\text{Var}(\hat{R}_1 + \hat{R}_2 + \hat{R}_3 - R_1 - R_2 - R_3) + 3\text{Var}(Q_2)\end{aligned}\tag{B.2}$$

Step 1: Bound on $\text{Var}(R_1 + R_2 + R_3)$ For notational simplicity, define

$$g_{i'} = K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'})$$

The following inequality holds with probability $1 - 3n^{-1/2}$ by Lemma B.2:

$$\begin{aligned}
\text{Var}(R_1) &\stackrel{(i)}{\leq} \frac{\text{Var}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} (g_{i'} - \mathbb{E}(g_{i'} \mid \zeta_i))\right)}{\underline{C}_1^2 n^2 h^{2d_\zeta}} \\
&\leq \frac{\left(\mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} (g_{i'} - \mathbb{E}(g_{i'} \mid \zeta_i))\right)\right)^2}{\underline{C}_1^2 n^2 h^{2d_\zeta}} \\
&\stackrel{(ii)}{\leq} \frac{\bar{C}_3^2 n h^{d_\zeta+4} \log n}{\underline{C}_1^2 n^2 h^{2d_\zeta}} \\
&= \frac{\bar{C}_3^2 h^4 \log n}{\underline{C}_1^2 n h^{d_\zeta}}
\end{aligned} \tag{B.3}$$

where (i) and (ii) come from (B.7) and (B.9) respectively. By the same argument, we have $\text{Var}(R_2) \leq \frac{\bar{C}_3^2 h^2 \log n}{\underline{C}_1^2 n h^{d_\zeta}}$ with probability at least $1 - 3n^{-1/2}$. For notational simplicity, define:

$$g_{i'j'} = K_{h,i'j'}(\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{i'j'}(\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)$$

For the bound on $\text{Var}(R_3)$, note that the following inequality holds with probability $1 - 3n^{-1/2}$ by Lemma B.3:

$$\begin{aligned}
\text{Var}(R_3) &\stackrel{(i)}{\leq} \frac{\text{Var}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} (g_{i'j'} - \mathbb{E}(g_{i'j'} \mid \zeta_i, \zeta_j))\right)}{\underline{D}_1^2 n^4 h^{4d_\zeta}} \\
&\leq \frac{\left(\mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} (g_{i'j'} - \mathbb{E}(g_{i'j'} \mid \zeta_i, \zeta_j))\right)\right)^2}{\underline{D}_1^2 n^4 h^{4d_\zeta}} \\
&\stackrel{(ii)}{\leq} \frac{\bar{D}_3^2 n^3 h^{3d_\zeta+4} \log n}{\underline{D}_1^2 n^4 h^{4d_\zeta}} \\
&= \frac{\bar{D}_3^2 h^4 \log n}{\underline{D}_1^2 n h^{d_\zeta}}
\end{aligned} \tag{B.4}$$

where (i) and (ii) come from (B.14) and (B.16) respectively.

Step 2: Bound on $\text{Var}(\hat{R}_1 + \hat{R}_2 + \hat{R}_3 - R_1 - R_2 - R_3)$ Since

$$\begin{aligned}
\text{Var}(\hat{R}_1 - R_1) &= \text{Var}\left(\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} - \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'}(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'}}\right) \\
&\stackrel{(i)}{\leq} \frac{1}{\min\{\underline{C}_1, \underline{C}_2\} n^2 h^{2d_\zeta}} \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(d_{ii'} \leq h \vee \hat{d}_{ii'} \leq h) (\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'}) (\hat{K}_{h,i'} - K_{h,i'})\right)^2 \\
&\stackrel{(ii)}{\leq} \frac{\bar{K}^2 (\delta_{N,n}/h)^2}{\min\{\underline{C}_1, \underline{C}_2\} n^2 h^{2d_\zeta}} \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(d_{ii'} \leq h \vee \hat{d}_{ii'} \leq h) (\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'})\right)^2 \\
&\stackrel{(iii)}{\leq} \frac{\bar{K}^2 (\delta_{N,n}/h)^2 \rho_{f''}^2 h^4 \max\{\eta^4, \lambda^{-4}\}}{\min\{\underline{C}_1, \underline{C}_2\} n^2 h^{2d_\zeta}} \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(d_{ii'} \leq h \vee \hat{d}_{ii'} \leq h)\right)^2 \\
&\leq \frac{2\bar{K}^2 (\delta_{N,n}/h)^2 \rho_{f''}^2 h^4 \max\{\eta^4, \lambda^{-4}\}}{\min\{\underline{C}_1, \underline{C}_2\} n^2 h^{2d_\zeta}} \left(\mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(d_{ii'} \leq h)\right)^2 + \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(\hat{d}_{ii'} \leq h)\right)^2\right)
\end{aligned}$$

where inequality (i) is from (B.7) and (B.8) in Lemma B.2, inequality (ii) holds because of the definition of $\mathcal{A}$, inequality (iii) follows from compact support of kernel function (Assumption 4) and Lemma B.1.

In addition, (B.7) and (B.8) in Lemma B.2 ensures that (by replacing K with uniform kernel on $[-1, 1]$) there exists constant $M > 0$ such that

$$\max \left\{ \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(d_{ii'} \leq h) \right)^2 + \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(\hat{d}_{ii'} \leq h) \right)^2 \right\} \leq M n^2 h^{2d_\zeta}$$

Thus, we must have

$$\text{Var}(\hat{R}_1 - R_1) \leq \frac{2M\bar{K}^2(\delta_{N,n}/h)^2 \rho_{f''}^2 h^4 \max\{\eta^4, \lambda^{-4}\}}{\min\{\underline{C}_1, \underline{C}_2\}} = \frac{2M\bar{K}^2 \delta_{N,n}^2 h^2 \rho_{f''}^2 \max\{\eta^4, \lambda^{-4}\}}{\min\{\underline{C}_1, \underline{C}_2\}} \quad (\text{B.5})$$

By the same argument, we can also show that $\text{Var}(\hat{R}_2 - R_2) = O(\delta_{N,n}^2 h^2)$ and $\text{Var}(\hat{R}_3 - R_3) = O(\delta_{N,n}^2 h^2)$.

Step 3: Bound on $\text{Var}(Q_2)$ Note that for $i, j \in \mathcal{S}^c$, (1) $\{\epsilon_{i'j'} \mid i' \in \mathcal{S}_2 \cup \{i\}, i' \in \mathcal{S}_2 \cup \{i\}\}$ is independent of $\{\hat{d}_{ii'} \mid i = 1, \dots, N, i' \in \mathcal{S}_2\}$ conditional on ζ (Assumption 1(ii)), (2) $\hat{K}_{ij}$ is uniformly bounded (Assumption 4), and (3) conditional ζ , $\{\epsilon_{i'j'} \mid i' \in \mathcal{S}_2 \cup \{i\}, i' \in \mathcal{S}_2 \cup \{i\}\}$ is a set of independent sub-Gaussian random variables with $\sigma = 1$, (4) $\text{Var}(\hat{K}_{h,i'\epsilon_{i'j}} \mid \zeta) = O(h^\zeta)$, $\text{Var}(\hat{K}_{h,i'j'\epsilon_{i'j'}} \mid \zeta) = O(h^{2d_\zeta})$, then we can employ Bernstein inequality to show there exists constant $\bar{M}$ such that for any $t_1, t_2 > 0$:

$$\begin{aligned} \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'\epsilon_{i'j}} \geq t \mid \zeta \right) &\leq \exp \left(\frac{-t^2}{\bar{M}^2 n h^{d_\zeta} + \frac{1}{3} \bar{M} t} \right) \\ \mathbb{P} \left(\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'\epsilon_{ij'}} \geq t \mid \zeta \right) &\leq \exp \left(\frac{-t^2}{\bar{M}^2 n h^{d_\zeta} + \frac{1}{3} \bar{M} t} \right) \\ \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'\epsilon_{i'j'}} \geq t \mid \zeta \right) &\leq \exp \left(\frac{-t^2}{\bar{M}^2 n^2 h^{2d_\zeta} + \frac{1}{3} \bar{M} t} \right) \end{aligned}$$

Then there exists $M_1, M_2 > 0$ such that

$$\begin{aligned} \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'\epsilon_{i'j}} \geq M_1 \sqrt{n h^{d_\zeta} \log n} \mid \zeta \right) &\leq n^{-1/2} \\ \mathbb{P} \left(\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j'\epsilon_{ij'}} \geq M_1 \sqrt{n h^{d_\zeta} \log n} \mid \zeta \right) &\leq n^{-1/2} \\ \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'\epsilon_{i'j'}} \geq M_2 n h^{d_\zeta} \sqrt{\log n} \mid \zeta \right) &\leq n^{-1/2} \end{aligned} \quad (\text{B.6})$$

We can show that with probability at least $1 - 3n^{-1/2}$:

$$\begin{aligned} \text{Var} \left(\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'\epsilon_{i'j}}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} \right) &\stackrel{(i)}{\leq} \frac{1}{\underline{C}_2^2 n^2 h^{2d_\zeta}} \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'\epsilon_{i'j}} \right)^2 \\ &\stackrel{(ii)}{\leq} \frac{M_1^2 n h^{d_\zeta} \log n}{\underline{C}_2^2 n^2 h^{2d_\zeta}} \\ &= \frac{M_1^2 \log n}{\underline{C}_2^2 n h^{d_\zeta}} \end{aligned}$$

where (i) holds because of (B.8) in Lemma B.2, and (ii) comes from (B.6). It directly follows that $\text{Var} \left(\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \epsilon_{i'j}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'}} \right) \leq \frac{M_1^2 \log n}{\underline{C}_2^2 n h^{d_\zeta}}$ with probability at least $1 - 3n^{-1/2}$. In addition, with probability at least $1 - 3n^{-1/2}$:

$$\begin{aligned} \text{Var} \left(\frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} \epsilon_{i'j'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'}} \right) &\stackrel{(i)}{\leq} \frac{1}{\underline{D}_2^2 n^4 h^{4d_\zeta}} \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} \epsilon_{i'j'} \right)^2 \\ &\stackrel{(ii)}{\leq} \frac{M_2^2 n^2 h^{2d_\zeta} \log n}{\underline{D}_2^2 n^4 h^{4d_\zeta}} \\ &= \frac{M_2^2 \log n}{\underline{D}_2^2 n^2 h^{2d_\zeta}} \end{aligned}$$

Thus, combining these results yields:

$$\text{Var} \left(\hat{A}_{ij} - \mathbb{E}(\hat{A}_{ij} \mid \zeta) \right) \leq O \left(\frac{h^4}{n h^{d_\zeta}} + \delta_{N,n}^2 h^2 + \frac{1}{n h^{d_\zeta}} + \frac{1}{n^2 h^{2d_\zeta}} \right)$$

By the asymptotics $n h^{d_\zeta} \rightarrow 0$, we have

$$\text{Var} \left(\hat{A}_{ij} - \mathbb{E}(\hat{A}_{ij} \mid \zeta) \right) \leq O \left(\delta_{N,n}^2 h^2 + \frac{1}{n h^{d_\zeta}} \right)$$

■

Lemma B.2 *Under conditions and notations in Theorem 2, for all $i, j \in \mathcal{S}^c$, we can find constants $0 < \underline{C}_1 < \bar{C}_1 < \infty$ and $0 < \underline{C}_2 < \bar{C}_2 < \infty$ such that:*

$$\mathbb{P} \left(\underline{C}_1 n h^{d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} \leq \bar{C}_1 n h^{d_\zeta} \right) \geq 1 - n^{-1/2} \quad (\text{B.7})$$

$$\mathbb{P} \left(\underline{C}_2 n h^{d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \leq \bar{C}_2 n h^{d_\zeta} \right) \geq 1 - 2n^{-1/2} \quad (\text{B.8})$$

In addition, there exists constant $\bar{C}_3 > 0$ such that:

$$\begin{aligned} &\mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} (K_{h,i'}(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'}) - \mathbb{E}(K_{h,i'}(\zeta_i - \zeta_{i'})' H_{i'}(\zeta_i - \zeta_{i'})) \mid \zeta_i) \leq \bar{C}_3 \sqrt{n h^{d_\zeta} + 4 \log n} \right) \\ &\geq 1 - n^{-1/2} \end{aligned} \quad (\text{B.9})$$

Proof of Lemma B.2. Note that for any fixed i ,

$$\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} - \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} \mid \zeta_i \right) = \sum_{i' \in \mathcal{S}_2} (K_{h,i'} - \mathbb{E}(K_{h,i'} \mid \zeta_i))$$

By the boundedness of $K_{h,i'}$ (Assumption 4) and $\text{Var}(K_{h,i'} - \mathbb{E}(K_{h,i'} \mid \zeta_i)) = O(h^{d_\zeta})$ (Assumption 1(iv), Assumption 4), Bernstein inequality of independent random variables (Wainwright (2019)) demonstrates

that there exists a constant $M > 0$ such that for any $t > 0$, we have:

$$\mathbb{P} \left(\sum_{i' \in \mathcal{S}_2} (K_{h,i'} - \mathbb{E}(K_{h,i'} \mid \zeta_i)) \geq t \mid \zeta_i \right) \leq 2 \exp \left(\frac{-t^2}{M^2 n h^{d_\zeta} + \frac{1}{3} M t} \right)$$

It follows that we can find a sufficiently large constant C_1 such that when $t = C_1 \sqrt{n h^{d_\zeta} \log n}$:

$$\mathbb{P} \left(\sum_{i' \in \mathcal{S}_2} (K_{h,i'} - \mathbb{E}(K_{h,i'} \mid \zeta_i)) \geq C_1 \sqrt{n h^{d_\zeta} \log n} \mid \zeta_i \right) \leq n^{-1/2} \quad (\text{B.10})$$

In addition,

$$\begin{aligned} \mathbb{E}(K_{h,i'} \mid \zeta_i) &= \int K \left(\frac{d(\zeta_i, \zeta_{i'})}{h} \right) d\mathbb{P}_\zeta(\zeta_{i'}) \\ &= \int_{\{\|\zeta_{i'} - \zeta_i\| \leq \lambda^{-1} h\}} K \left(\frac{d(\zeta_i, \zeta_{i'})}{h} \right) d\mathbb{P}_\zeta(\zeta_{i'}) \\ &\stackrel{(i)}{\geq} C_2 \mathbb{P}(\|\zeta_i - \zeta_{i'}\| \leq \lambda^{-1} h \mid \zeta_i) \\ &\stackrel{(ii)}{\geq} C_2 \underline{c} (\lambda^{-1} h)^{d_\zeta} \end{aligned} \quad (\text{B.11})$$

Inequality (i) follows from the Lipschitz continuity of kernel K and $K(0) > 0$ (see Assumption 4), and inequality (ii) holds because Assumption 1(iv). By the boundedness of kernel K we can also show that there exists constant $C_3 > 0$ such that

$$\mathbb{E}(K_{h,i'} \mid \zeta_i) \leq C_3 \bar{c} (\lambda^{-1} h)^{d_\zeta} \quad (\text{B.12})$$

Combining (B.10), (B.11), and (B.12), we have

$$\mathbb{P} \left(C_2 \underline{c} n (\lambda^{-1} h)^{d_\zeta} - C_1 \sqrt{n h^{d_\zeta} \log n} \leq \sum_{i' \in \mathcal{S}_2} K_{h,i'} \leq C_3 \bar{c} n (\lambda^{-1} h)^{d_\zeta} + C_1 \sqrt{n h^{d_\zeta} \log n} \mid \zeta_i \right) \leq 1 - n^{-1/2}$$

Therefore, we can find constants $0 < \underline{C}_1 < \bar{C}_1 < \infty$ such that

$$\mathbb{P} \left(\underline{C}_1 n h^{d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} \leq \bar{C}_1 n h^{d_\zeta} \mid \zeta_i \right) \geq 1 - n^{-1/2}$$

Since the inequality above holds for any ζ_i , we can drop the conditioning and obtain (B.7).

Now let us turn to the derivation of (B.8). Note that when conditional on event $\mathcal{A}$, we have

$$\begin{aligned} \left| \sum_{i' \in \mathcal{S}_2 \cup \{i\}} (\hat{K}_{h,i'} - K_{h,i'}) \right| &\stackrel{(i)}{\leq} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(\hat{d}_{ii'} \leq h) \bar{K} |\hat{d}_{ii'} - d_{ii'}|/h \\ &\stackrel{(ii)}{\leq} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(\hat{d}_{ii'} \leq h) \bar{K} \delta_{N,n}/h \end{aligned}$$

where inequality (i) holds because of Assumption 4, and inequality (ii) comes from the definition of $\mathcal{A}$ in as in (B.1). Since we require $\delta_{N,n}/n \rightarrow 0$, we can employ the same argument used in proving (B.7) to

show that there exists constants $0 < C_4 < C_5 < \infty$ such that

$$\begin{aligned} & \mathbb{P} \left(C_4 n h^{d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbf{1}(\hat{d}_{ii'} \leq h) \leq C_5 n h^{d_\zeta} \mid \mathcal{A} \right) \leq 1 - n^{-1/2} \\ \Rightarrow & \mathbb{P} \left(C_4 n h^{d_\zeta - 1} \delta_{N,n} \leq \left| \sum_{i' \in \mathcal{S}_2 \cup \{i\}} (\hat{K}_{h,i'} - K_{h,i'}) \right| \leq C_5 n h^{d_\zeta - 1} \delta_{N,n} \mid \mathcal{A} \right) \leq 1 - n^{-1/2} \end{aligned} \quad (\text{B.13})$$

Then combining (B.7) and (B.13), we can find $0 < \underline{C}_2 < \bar{C}_2 < \infty$ such that:

$$\mathbb{P} \left(\underline{C}_2 n h^{d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i'} \leq \bar{C}_2 n h^{d_\zeta} \mid \mathcal{A} \right) \geq 1 - n^{-1/2}$$

Therefore, we can obtain (B.8) by Lemma 1.

To prove (B.9), note that for any fixed i ,

$$\begin{aligned} & \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) - \mathbb{E} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i \right) \\ &= \sum_{i' \in \mathcal{S}_2} (K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) - \mathbb{E}(K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i)) \end{aligned}$$

By the boundedness of $K_{h,i'}$ and $H_{i'}$ (Assumption 4 and Assumption 1(v)) and

$$\text{Var}(K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) - \mathbb{E}(K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i)) = O(h^{d_\zeta + 4})$$

(see Assumption 1(iv), Assumption 4), Bernstein inequality of independent random variables (Wainwright (2019)) demonstrates that there exists a constant $M > 0$ such that for any $t > 0$, we have:

$$\begin{aligned} & \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) - \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbb{E}(K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i) \geq t \mid \zeta_i \right) \\ & \leq 2 \exp \left(\frac{-t^2}{M^2 n h^{d_\zeta + 4} + \frac{1}{3} M t} \right) \end{aligned}$$

It follows that we can find a sufficiently large constant $\bar{C}_3$ such that when $t = \bar{C}_3 \sqrt{n h^{d_\zeta + 4} \log n}$:

$$\begin{aligned} & \mathbb{P} \left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) - \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \mathbb{E}(K_{h,i'} (\zeta_i - \zeta_{i'})' H_{i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i) \geq \bar{C}_3 \sqrt{n h^{d_\zeta + 4} \log n} \mid \zeta_i \right) \\ & \leq n^{-1/2} \end{aligned}$$

Since the inequality above is true for any ζ_i , we can drop the conditioning and obtain (B.9). ■

Lemma B.3 *Under conditions and notations in Theorem 2, for all $i, j \in \mathcal{S}^c$, we can find constants $0 < \underline{D}_1 < \bar{D}_1 < \infty$ and $0 < \underline{D}_2 < \bar{D}_2 < \infty$ such that:*

$$\mathbb{P} \left(\underline{D}_1 n^2 h^{2d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,i'j'} \leq \bar{D}_1 n^2 h^{2d_\zeta} \right) \geq 1 - n^{-1/2} \quad (\text{B.14})$$

$$\mathbb{P}\left(\underline{D}_2 n^2 h^{2d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,i'j'} \leq \bar{D}_2 n^2 h^{2d_\zeta}\right) \geq 1 - 2n^{-1/2} \quad (\text{B.15})$$

In addition, there exists constant $\bar{D}_3 > 0$ such that:

$$\begin{aligned} & \mathbb{P}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} (K_{h,i'j'}(\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'})' H_{i'j'}(\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'})) \right. \\ & \quad \left. - \mathbb{E}(K_{h,i'j'}(\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'})' H_{i'j'}(\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'}) \mid \zeta_i, \zeta_j) \leq \bar{D}_3 \sqrt{n^3 h^{3d_\zeta + 4} \log n}\right) \geq 1 - n^{-1/2} \end{aligned} \quad (\text{B.16})$$

Proof of Lemma B.3. Consider the decomposition:

$$\begin{aligned} & \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,i'j'} \mid \zeta_i, \zeta_j\right) \\ &= K(0) \left(\sum_{i' \in \mathcal{S}_2} K_{h,i'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2} K_{h,i'} \mid \zeta_i\right)\right) + K(0) \left(\sum_{j' \in \mathcal{S}_2} K_{h,j'} - \mathbb{E}\left(\sum_{j' \in \mathcal{S}_2} K_{h,j'} \mid \zeta_j\right)\right) \\ & \quad + \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} \left(K_{h,i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h,i'j'} \mid \zeta_i, \zeta_j\right)\right) \end{aligned} \quad (\text{B.17})$$

The property of the first two terms has been developed in Lemma B.2, so we focus on the third term in the following proof. Note that $K_{h,i'j'}$ is uniformly bounded and $\mathbb{E}(\mathbb{E}^2(K_{h,i'j'} \mid \zeta_i, \zeta_j, \zeta_{i'}) \mid \zeta_i, \zeta_j) = O(h^{3d_\zeta})$, then by Bernstein's inequality of U-statistics (Theorem 2 in Arcones (1995)), there exists sufficiently large constant M such that for any $t > 0$, we have

$$\mathbb{P}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h,i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h,i'j'} \mid \zeta_i, \zeta_j\right) \geq t\right) \leq 4 \exp\left(\frac{-t^2}{M^2 n^3 h^{3d_\zeta} + Mnt}\right)$$

We can find a constant D_1 such that when $t = D_1 \sqrt{n^3 h^{3d_\zeta} \log n}$,

$$\mathbb{P}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h,i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h,i'j'} \mid \zeta_i, \zeta_j\right) \geq D_1 \sqrt{n^3 h^{3d_\zeta} \log n}\right) \leq n^{-1/2} \quad (\text{B.18})$$

In addition,

$$\begin{aligned} \mathbb{E}(w_{i'j'} \mid \zeta_i, \zeta_j) &= \int K\left(\frac{d(\zeta_i, \zeta_{i'})}{h}\right) K\left(\frac{d(\zeta_j, \zeta_{j'})}{h}\right) d\mathbb{P}_\zeta(\zeta_{i'}) d\mathbb{P}_\zeta(\zeta_{j'}) \\ &= \int_{\{\|\zeta_{i'} - \zeta_i\| \leq \lambda^{-1}h\}} K\left(\frac{d(\zeta_i, \zeta_{i'})}{h}\right) d\mathbb{P}_\zeta(\zeta_{i'}) \int_{\{\|\zeta_{j'} - \zeta_j\| \leq \lambda^{-1}h\}} K\left(\frac{d(\zeta_j, \zeta_{j'})}{h}\right) d\mathbb{P}_\zeta(\zeta_{j'}) \\ &\stackrel{(i)}{\geq} D_2 \mathbb{P}(\|\zeta_i - \zeta_{i'}\| \leq \lambda^{-1}h \mid \zeta_i) \mathbb{P}(\|\zeta_j - \zeta_{j'}\| \leq \lambda^{-1}h \mid \zeta_j) \\ &\stackrel{(ii)}{\geq} D_2 \underline{c}^2 (\lambda^{-1}h)^{2d_\zeta} \end{aligned} \quad (\text{B.19})$$

Inequality (i) follows from the Lipschitz continuity of kernel K and $K(0) > 0$ (see Assumption 4), and inequality (ii) holds because Assumption 1(iv). By the boundedness of kernel K we can also show that there exists constant $D_3 > 0$ such that

$$\mathbb{E}(K_{h,i'j'} \mid \zeta_i) \leq D_3 \bar{c}^2 (\lambda^{-1}h)^{2d_\zeta} \quad (\text{B.20})$$

Combining (B.18), (B.19), and (B.20), we have

$$\begin{aligned} & \mathbb{P} \left(D_2 \bar{c}^2 n^2 (\lambda^{-1} h)^{2d_\zeta} - D_1 \sqrt{n^3 h^{3d_\zeta} \log n} \leq \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} K_{h, i'} \leq D_3 \bar{c}^2 n^2 (\lambda^{-1} h)^{2d_\zeta} + C_1 \sqrt{n^3 h^{3d_\zeta} \log n} \mid \zeta_i, \zeta_j \right) \\ & \leq 1 - n^{-1/2} \end{aligned}$$

In addition, by Lemma B.2, it is straightforward to see that the order of the first two terms in (B.17) is asymptotically negligible compared to the third term. Thus, we can find constants $0 < \underline{D}_1 < \bar{D}_1 < \infty$ such that

$$\mathbb{P} \left(\underline{D}_1 n^2 h^{2d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h, i' j'} \leq \bar{D}_1 n^2 h^{2d_\zeta} \mid \zeta_i, \zeta_j \right) \geq 1 - n^{-1/2}$$

Since the inequality above holds for any ζ_i, ζ_j , we can drop the conditioning and obtain (B.14).

Now let us turn to the derivation of (B.15). Note that when conditional on event $\mathcal{A}$, we have

$$\begin{aligned} \left| \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} (\hat{K}_{h, i' j'} - K_{h, i' j'}) \right| & \stackrel{(i)}{\leq} \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} \mathbf{1}(\hat{d}_{ii'} \leq h) \mathbf{1}(\hat{d}_{jj'} \leq h) \bar{K} \rho_K \frac{|\hat{d}_{ii'} - d_{ii'}| + |\hat{d}_{jj'} - d_{jj'}|}{h} \\ & \quad + \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} \mathbf{1}(\hat{d}_{ii'} \leq h) \mathbf{1}(\hat{d}_{jj'} \leq h) \bar{K}^2 \frac{|\hat{d}_{ii'} - d_{ii'}| |\hat{d}_{jj'} - d_{jj'}|}{h^2} \\ & \stackrel{(ii)}{\leq} \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} \mathbf{1}(\hat{d}_{ii'} \leq h) \mathbf{1}(\hat{d}_{jj'} \leq h) \left(\bar{K} \rho_K \frac{\delta_{N,n}}{h} + \bar{K}^2 \frac{\delta_{N,n}^2}{h^2} \right) \end{aligned}$$

where inequality (i) holds because of Assumption 4, and inequality (ii) comes from the definition of $\mathcal{A}$ in as in (B.1). Since we require $\delta_{N,n}/n \rightarrow 0$, we can employ the same argument used in proving (B.7) to show that there exists constants $0 < D_4 < D_5 < \infty$ such that

$$\begin{aligned} & \mathbb{P} \left(D_4 n^2 h^{2d_\zeta} \leq \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} \mathbf{1}(\hat{d}_{ii'} \leq h) \mathbf{1}(\hat{d}_{jj'} \leq h) \leq D_5 n^2 h^{2d_\zeta} \mid \mathcal{A} \right) \leq 1 - n^{-1/2} \\ \Rightarrow & \mathbb{P} \left(D_4 n^2 h^{2d_\zeta - 1} \delta_{N,n} \leq \left| \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} (\hat{K}_{h, i' j'} - K_{h, i' j'}) \right| \leq D_5 n^2 h^{2d_\zeta - 1} \delta_{N,n} \mid \mathcal{A} \right) \leq 1 - n^{-1/2} \end{aligned} \tag{B.21}$$

Then combining (B.14) and (B.21), we can find $0 < \underline{D}_2 < \bar{D}_2 < \infty$ such that:

$$\mathbb{P} \left(\underline{D}_2 n^2 h^{2d_\zeta} \leq \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h, i' j'} \leq \bar{D}_2 n^2 h^{2d_\zeta} \mid \mathcal{A} \right) \geq 1 - n^{-1/2}$$

Therefore, we can obtain (B.15) by Lemma 1.

Let $g_{i' j'}$ to denote $K_{h, i' j'} (\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'})' H_{i' j'} (\zeta_i - \zeta_{i'}, \zeta_j - \zeta_{j'})$ for notational simplicity. To prove (B.16), note that

$$\begin{aligned} & \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} (g_{i' j'} - \mathbb{E}(g_{i' j'} \mid \zeta_i, \zeta_j)) \\ & = \sum_{i' \in \mathcal{S}_2} (g_{i' j} - \mathbb{E}(g_{i' j} \mid \zeta_i, \zeta_j)) + \sum_{j' \in \mathcal{S}_2} (g_{i j'} - \mathbb{E}(g_{i j'} \mid \zeta_i, \zeta_j)) + \sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} (g_{i' j'} - \mathbb{E}(g_{i' j'} \mid \zeta_i, \zeta_j)) \end{aligned} \tag{B.22}$$

The property of the first two terms has been developed in Lemma B.2, so we focus on the third term in the following proof. Note that by the boundedness of $K_{h, i'}$ and $H_{i'}$ (Assumption 4 and Assump-

tion 1(v)), we have that $g_{i'j'}$ is uniformly bounded and $\mathbb{E}(\mathbb{E}^2(g_{h,i'j'} \mid \zeta_i, \zeta_j, \zeta_{i'}) \mid \zeta_i, \zeta_j) = O(h^{3d_\zeta+4})$. Then by Bernstein's inequality of U-statistics (Theorem 2 in Arcones (1995)), there exists sufficiently large constant M such that for any $t > 0$, we have

$$\mathbb{P}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} g_{i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} g_{i'j'} \mid \zeta_i, \zeta_j\right) \geq t\right) \leq 4 \exp\left(\frac{-t^2}{M^2 n^3 h^{3d_\zeta+4} + M n h^2 t}\right)$$

It follows that we can find a sufficiently large constant D_5 such that when $t = D_5 \sqrt{n^3 h^{3d_\zeta+4} \log n}$:

$$\mathbb{P}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} g_{i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2, j' \in \mathcal{S}_2} g_{i'j'} \mid \zeta_i, \zeta_j\right) \geq D_5 \sqrt{n^3 h^{3d_\zeta+4} \log n}\right) \leq n^{-1/2}$$

In addition, by Lemma B.2, it is straightforward to see that the order of the first two terms in (B.22) is asymptotically negligible compared to the third term. Thus, we can find constants $\bar{D}_3 > 0$ such that

$$\mathbb{P}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} g_{i'j'} - \mathbb{E}\left(\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} g_{i'j'} \mid \zeta_i, \zeta_j\right) \geq \bar{D}_3 \sqrt{n^3 h^{3d_\zeta+4} \log n}\right) \leq n^{-1/2}$$

Since the inequality above is true for any ζ_i, ζ_j , we can drop the conditioning and obtain (B.16). ■

C Proofs for Empirical Applications

Proof of Theorem 3.

$$\begin{aligned} \sup_{\beta} \left\| \psi(\hat{A}, X_m, Y_m, Y_m, \beta) - \mu(\beta) \right\| &\leq \sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(\hat{A}, X_m, Y_m, Y_m, \beta) - \frac{1}{M} \sum_{m=1}^M \psi(F_m, X_m, Y_m, Y_m, \beta) \right\| \\ &\quad + \sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(F_m, X_m, Y_m, Y_m, \beta) - \frac{1}{M} \sum_{m=1}^M \psi(A_m, X_m, Y_m, Y_m, \beta) \right\| \\ &\quad + \sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(A_m, X_m, Y_m, Y_m, \beta) - \mathbb{E}(\psi(A_m, X_m, W_m, Y_m, \beta)) \right\| \end{aligned}$$

Note that

$$\begin{aligned} &\sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(\hat{A}, X_m, Y_m, Y_m, \beta) - \frac{1}{M} \sum_{m=1}^M \psi(F_m, X_m, Y_m, Y_m, \beta) \right\| \\ &\leq \frac{1}{M} \sum_{m=1}^M L(X_m, W_m, Y_m) N^{-1} \|\hat{F}_{m1} - F_{m2}\|_F \\ &\stackrel{(i)}{=} o_P(1) \end{aligned}$$

where (i) is from Assumption 5(vii) and Assumption 5(viii). In addition, by Assumption 5(iv), we have

$$\sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(F_m, X_m, Y_m, Y_m, \beta) - \frac{1}{M} \sum_{m=1}^M \psi(A_m, X_m, Y_m, Y_m, \beta) \right\| = o_P(1)$$

By Assumption 5??, we have

$$\sup_{\beta} \left\| \frac{1}{M} \sum_{m=1}^M \psi(A_m, X_m, Y_m, Y_m, \beta) - \mathbb{E}(\psi(A_m, X_m, W_m, Y_m, \beta_0)) \right\| = o_P(1)$$

Thus, we must have $\sup_{\beta} \left\| \psi(\hat{A}, X_m, Y_m, Y_m, \beta) - \mathbb{E}(\psi(A_m, X_m, W_m, Y_m, \beta_0)) \right\| \xrightarrow{p} 0$. Combining the uniform convergence result above, Assumption 5(iii), (v), ??, and Theorem 2.1 in Newey and McFadden (1994), we can prove that $\hat{\beta} \xrightarrow{p} \beta_0$. ■

For notational simplicity, define

$$p(\zeta) = \int f(\zeta, \zeta') d\mathbb{P}(\zeta')$$

and assume that $\inf_{\zeta} p(\zeta) > 0$.

Theorem C.1 1

Proof of Theorem C.1. Consider the linear regression model:

$$Y = \beta GW + e$$

which is a simple linear-mean regression model, and let us consider the asymptotics of GW and therefore, $\hat{\beta}$. We always assume that W is fixed or conditional on W , e are independent with each other. The OLS

estimator is

$$\begin{aligned}\hat{\beta} &= \left(W' \hat{G}' \hat{G} W\right)^{-1} W' \hat{G}' Y \\ &= \beta - \beta \left(W' \hat{G}' \hat{G} W\right)^{-1} W' \hat{G}' (\hat{G} - G) W + \left(W' \hat{G}' \hat{G} W\right)^{-1} W' (\hat{G} - G)' e + \left(W' \hat{G}' \hat{G} W\right)^{-1} W' G' e\end{aligned}$$

Note that for each $i \in \mathcal{S}^c$, we have

$$\begin{aligned}(\hat{G}W - GW)_i &= \frac{\sum_{j=1,\dots,N} A_{ij} W_j + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij}) W_j}{\sum_{j=1,\dots,N} A_{ij} + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij})} - \frac{\sum_{j=1,\dots,N} A_{ij} W_j}{\sum_{j=1,\dots,N} A_{ij}} \\ &= \underbrace{\left(\sum_{j=1,\dots,N} A_{ij} W_j \right) \left(\frac{1}{\sum_{j=1,\dots,N} A_{ij} + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij})} - \frac{1}{\sum_{j=1,\dots,N} A_{ij}} \right)}_{B_1} \\ &\quad + \underbrace{\frac{\sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij}) W_j}{\sum_{j=1,\dots,N} A_{ij} + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij})}}_{B_2}\end{aligned}$$

By Lemma C.2, we have

$$\begin{aligned}\mathbb{E} \left(\frac{1}{\sum_{j=1,\dots,N} A_{ij} + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij})} - \frac{1}{\sum_{j=1,\dots,N} A_{ij}} \right) &= \frac{1}{N} O \left(h^2 + h^4 + \frac{h^4}{nh^{d_\zeta}} + h^2 \delta_{N,n}^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right) \\ &= \frac{1}{N} O \left(h^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right)\end{aligned}$$

Since $\sum_{j=1,\dots,N} A_{ij} W_j = O(N)$, we have:

$$\mathbb{E}(B_1) = O \left(h^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right)$$

In addition,

$$\begin{aligned}\text{Var} \left(\frac{1}{\sum_{j=1,\dots,N} A_{ij} + \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij})} - \frac{1}{\sum_{j=1,\dots,N} A_{ij}} \right) \\ \mathbb{E} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - p(\zeta_i)) \right) = 0, \quad \text{Var} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - p(\zeta_i)) \right) = O(N)\end{aligned}$$

$$\hat{G}_{ij} - G_{ij} = O(h^2) + \frac{1}{nh^d} \sum_{j' \in \mathcal{N}_j} \epsilon_{ij'} + \frac{1}{nh^d} \sum_{i' \in \mathcal{N}_i} \epsilon_{i'j} - \frac{1}{n^2 h^{2d}} \sum_{i' \in \mathcal{N}_i, j' \in \mathcal{N}_j} \epsilon_{i'j'}$$

■

Lemma C.2 Under conditions in Theorem 2, we have that for any $i \in \mathcal{S}^c$:

$$\mathbb{E} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - \hat{A}_{ij}) \right) = O(Nh^2), \quad \text{Var} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - \hat{A}_{ij}) \right) = N^2 O \left(\frac{h^4}{nh^{d_\zeta}} + h^2 \delta_{N,n}^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right) \quad (\text{C.1})$$

In addition, there exist constant $\bar{E}_1 > 0$ and $c > 0$, such that:

$$\mathbb{P} \left(\left| \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij}) \right| \leq \bar{E}_1 N \left(h^2 + \frac{h^2}{\sqrt{nh^{d_\zeta}}} + h\delta_{N,n} + \frac{1}{\sqrt{n}} + \frac{1}{nh^{d_\zeta}} \right) \right) \geq 1 - cn^{-1/2} \quad (\text{C.2})$$

In addition, when $\{W_i \mid i = 1, 2, \dots, N\}$ are uniformly bounded and $\mathbb{E}(\epsilon_{ij} \mid \{W_i \mid i = 1, 2, \dots, N\}) = 0$ for any i, j , we have:

$$\mathbb{E} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - \hat{A}_{ij}) W_j \right) = O(Nh^2), \quad \text{Var} \left(\sum_{j \in \mathcal{S}^c} (A_{ij} - \hat{A}_{ij}) W_j \right) = N^2 O \left(\frac{h^4}{nh^{d_\zeta}} + h^2 \delta_{N,n}^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right) \quad (\text{C.3})$$

In addition, there exist constant $\bar{E}_2 > 0$ and $c > 0$, such that:

$$\mathbb{P} \left(\left| \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij}) W_j \right| \leq \bar{E}_2 N \left(h^2 + \frac{h^2}{\sqrt{nh^{d_\zeta}}} + h\delta_{N,n} + \frac{1}{\sqrt{n}} + \frac{1}{nh^{d_\zeta}} \right) \right) \geq 1 - cn^{-1/2} \quad (\text{C.4})$$

Proof of Lemma C.2. For notational simplicity, we define

$$\begin{aligned} K_{h,i,i'} &:= K(d_{ii'}/h), \quad K_{h,j,j'} := K(d_{jj'}/h), \quad K_{h,ij,i'j'} := K_{h,i,i'} K_{h,j,j'} \\ \hat{K}_{h,i,i'} &:= K(\hat{d}_{ii'}/h), \quad \hat{K}_{h,j,j'} := K(\hat{d}_{jj'}/h), \quad \hat{K}_{h,ij,i'j'} := \hat{K}_{h,i,i'} \hat{K}_{h,j,j'} \end{aligned}$$

Note that for any $i, j \in \mathcal{S}^c$,

$$\hat{A}_{ij} - A_{ij} = R_{ij,1} + R_{ij,2} + R_{ij,3} + (\hat{R}_{ij,1} - R_{ij,1} + \hat{R}_{ij,2} - R_{ij,2} + \hat{R}_{ij,3} - R_{ij,3}) + Q_{ij} + \epsilon_{ij}$$

where

$$\begin{aligned} R_{ij,1} &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}}, \quad R_{ij,2} = \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j,j'} (\zeta_j - \zeta_{j'})' H_{j,j'} (\zeta_j - \zeta_{j'})}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'}}, \\ R_{ij,3} &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'}} \\ \hat{R}_{ij,1} &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}}, \quad \hat{R}_{ij,2} = \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'} (\zeta_j - \zeta_{j'})' H_{j,j'} (\zeta_j - \zeta_{j'})}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'}}, \\ \hat{R}_{ij,3} &= \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'}} \end{aligned}$$

and

$$Q_{ij} = \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'} \epsilon_{ij}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}} + \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'} \epsilon_{ij}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'}} - \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'} \epsilon_{ij}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'}}$$

then

$$\begin{aligned} \sum_{j \in 1, \dots, N} (\hat{A}_{ij} - A_{ij}) &= \sum_{j \in \mathcal{S}^c} (R_{ij,1} + R_{ij,2} + R_{ij,3}) \\ &\quad + \sum_{j \in \mathcal{S}^c} (\hat{R}_{ij,1} - R_{ij,1} + \hat{R}_{ij,2} - R_{ij,2} + \hat{R}_{ij,3} - R_{ij,3}) + \sum_{j \in \mathcal{S}^c} Q_{ij} + \sum_{j \in \mathcal{S}^c} \epsilon_{ij} \end{aligned}$$

The bias of $\sum_{j \in 1, \dots, N} (\hat{A}_{ij} - A_{ij})$ is of order at most (Nh^2) , and we focus on the variance of $\sum_{j \in 1, \dots, N} (\hat{A}_{ij} - A_{ij})$ in the following text.

Step 1: $\sum_{j \in \mathcal{S}^c} (R_{ij,1} + R_{ij,2} + R_{ij,3})$ Recall that

$$\sum_{j \in \mathcal{S}^c} R_{ij,1} = \sum_{j \in \mathcal{S}^c} \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}}$$

Note that since

$$\begin{aligned} \mathbb{E} (\mathbb{E}^2 (K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i, \zeta_j) \mid \zeta_i) &= O(h^{2d_\zeta+4}) \\ \mathbb{E} (\mathbb{E}^2 (K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i, \zeta_{i'}) \mid \zeta_i) &= O(h^{d_\zeta+4}) \end{aligned}$$

by Hoeffding decomposition, there exist large constant M such that for any $t > 0$, we have:

$$\begin{aligned} &\mathbb{P} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) - \mathbb{E} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i \right) \geq t \mid \zeta_i \right) \\ &\leq 4 \exp \left(\frac{-t^2}{M^2 n^2 N h^{2d_\zeta+4} + M n h^2 t} \right) + 4 \exp \left(\frac{-t^2}{M^2 n N^2 h^{d_\zeta+4} + M N h^2 t} \right) \end{aligned}$$

Then there exist a constant $E_1 > 0$ such that when $t = E_1 \sqrt{n N^2 h^{d_\zeta+4} \log n}$, we have:

$$\begin{aligned} &\mathbb{P} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) - \mathbb{E} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'}) \mid \zeta_i \right) \right. \\ &\quad \left. \geq E_1 \sqrt{n N^2 h^{d_\zeta+4} \log n} \mid \zeta_i \right) \leq 1 - n^{1/2} \end{aligned} \tag{C.5}$$

By the same method, we can also show that there exist constant E_2, E_3 such that:

$$\begin{aligned} &\mathbb{P} \left(\sum_{j \in \mathcal{S}^c} \sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j,j'} (\zeta_j - \zeta_{j'})' H_{j,j'} (\zeta_j - \zeta_{j'}) - \mathbb{E} \left(\sum_{j \in \mathcal{S}^c} \sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j,j'} (\zeta_j - \zeta_{j'})' H_{j,j'} (\zeta_j - \zeta_{j'}) \mid \zeta_i \right) \right. \\ &\quad \left. \geq E_2 \sqrt{n N^2 h^{3d_\zeta+4} \log n} \right) \leq 1 - n^{1/2} \end{aligned} \tag{C.6}$$

and

$$\begin{aligned} &\mathbb{P} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j) \right. \\ &\quad \left. - \mathbb{E} \left(\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j) \mid \zeta_i \right) \right. \\ &\quad \left. \geq E_3 \sqrt{n^3 N^2 h^{5d_\zeta+4} \log n} \right) \leq 1 - n^{1/2} \end{aligned} \tag{C.7}$$

Thus, by (C.5), (C.6), (C.7), (B.8), and (B.15), we have

$$\begin{aligned}\text{Var} \left(\sum_{j \in \mathcal{S}^c} R_{ij,1} \right) &= \text{Var} \left(\sum_{j \in \mathcal{S}^c} \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} K_{h,i,i'} (\zeta_i - \zeta_{i'})' H_{i,i'} (\zeta_i - \zeta_{i'})}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}} \right) \\ &= O \left(\frac{1}{n^2 h^{2d_\zeta}} \right) O \left(N^2 (n d^{\zeta+4} \log n) \right) \\ &= O \left(\frac{N^2 h^4 \log n}{n h^{d_\zeta}} \right)\end{aligned}\tag{C.8}$$

$$\begin{aligned}\text{Var} \left(\sum_{j \in \mathcal{S}^c} R_{ij,2} \right) &= \text{Var} \left(\sum_{j \in \mathcal{S}^c} \frac{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} K_{h,j,j'} (\zeta_j - \zeta_{j'})' H_{j,j'} (\zeta_j - \zeta_{j'})}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'}} \right) \\ &= O \left(\frac{1}{n^2 h^{2d_\zeta}} \right) O \left(N^2 n h^{3d_\zeta+4} \log n \right) \\ &= \frac{N^2 h^{d_\zeta+4} \log n}{n}\end{aligned}\tag{C.9}$$

$$\begin{aligned}\text{Var} \left(\sum_{j \in \mathcal{S}^c} R_{ij,3} \right) &= \text{Var} \left(\sum_{j \in \mathcal{S}^c} \frac{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} K_{h,ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)' H_{ij,i'j'} (\zeta_{i'} - \zeta_i, \zeta_{j'} - \zeta_j)}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'}} \right) \\ &= O \left(\frac{1}{n^4 h^{4d_\zeta}} \right) O \left(n^3 N^2 h^{5d_\zeta+4} \log n \right) \\ &= \frac{N^2 h^{d_\zeta+4} \log n}{n}\end{aligned}\tag{C.10}$$

Step 2: $\sum_{j \in \mathcal{S}^c} (\hat{R}_{ij,1} - R_{ij,1} + \hat{R}_{ij,2} - R_{ij,2} + \hat{R}_{ij,3} - R_{ij,3})$ It is easy to check that by Lemma 1 and using the same argument in the proof of Theorem 2, we can show that there exist a constant $E_4 > 0$ such that with probability at least $1 - n^{-1/2}$:

$$\sum_{j \in \mathcal{S}^c} (\hat{R}_{ij,1} - R_{ij,1} + \hat{R}_{ij,2} - R_{ij,2} + \hat{R}_{ij,3} - R_{ij,3}) \leq E_4 N h \delta_{N,n}\tag{C.11}$$

$$\text{Var} \left(\sum_{j \in \mathcal{S}^c} (\hat{R}_{ij,1} - R_{ij,1} + \hat{R}_{ij,2} - R_{ij,2} + \hat{R}_{ij,3} - R_{ij,3}) \right) = O \left(N^2 h^2 \delta_{N,n}^2 \right)\tag{C.12}$$

Step 3: $\sum_{j \in \mathcal{S}^c} (Q_{ij})$ Note that by the conditional independence between $\{\hat{d}_{ii'} \mid i \in \mathcal{S}_2, i' \in \mathcal{S}_2 \cup \mathcal{S}^c\}$ and $\{\epsilon_{ij} \mid i < j, i, j \in \mathcal{S}_2 \cup \mathcal{S}^c\}$, by Hoeffding inequality and the same method used in Theorem 2, we immediately have that there exist constant $E_5, E_6, E_7 > 0$ such that with probability at least $1 - n^{-1/2}$:

$$\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i' \epsilon_{ij}} \leq E_5 \sqrt{N n h^{d_\zeta} \log N}\tag{C.13}$$

$$\sum_{j \in \mathcal{S}^c} \sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j' \epsilon_{ij'}} = \sum_{j' \in \mathcal{S}_2} \left(\sum_{j \in \mathcal{S}^c} \hat{K}_{h,j,j'} \right) \epsilon_{ij'} + K(0) \sum_{j \in \mathcal{S}^c} \epsilon_{ij} \leq E_6 \sqrt{N^2 n h^{2d_\zeta} \log N}\tag{C.14}$$

$$\begin{aligned}\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j' \epsilon_{ij'}} &= \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2} \left(\sum_{j \in \mathcal{S}^c} \hat{K}_{h,ij,i'j'} \right) \epsilon_{i'j'} + K(0) \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \sum_{j \in \mathcal{S}^c} \hat{K}_{h,i,i' \epsilon_{ij}} \\ &= E_7 \sqrt{n^2 N^2 h^{2d_\zeta} \log N}\end{aligned}\tag{C.15}$$

Thus, by (C.13), (C.14), (C.15), (B.8), and (B.15), we have

$$\text{Var} \left(\frac{\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'} \epsilon_{i'j}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}} \hat{K}_{h,i,i'}} \right) = O \left(\frac{N n h^{d_\zeta} \log N}{n^2 h^{2d_\zeta}} \right) = O \left(\frac{N}{n h^{d_\zeta}} \right) \quad (\text{C.16})$$

$$\text{Var} \left(\frac{\sum_{j \in \mathcal{S}^c} \sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'} \epsilon_{ij'}}{\sum_{j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,j,j'}} \right) = O \left(\frac{N^2 n h^{2d_\zeta} \log N}{n^2 h^{2d_\zeta}} \right) = O \left(\frac{N^2}{n} \right) \quad (\text{C.17})$$

$$\text{Var} \left(\frac{\sum_{j \in \mathcal{S}^c} \sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'} \epsilon_{i'j'}}{\sum_{i' \in \mathcal{S}_2 \cup \{i\}, j' \in \mathcal{S}_2 \cup \{j\}} \hat{K}_{h,ij,i'j'}} \right) = O \left(\frac{N^2 n^2 h^{2d_\zeta} \log N}{n^4 h^{4d_\zeta}} \right) = O \left(\frac{N^2 \log N}{n^2 h^{2d_\zeta}} \right) \quad (\text{C.18})$$

Step 4 $\sum_{j \in \mathcal{S}^c} \epsilon_{ij}$ By the conditional independence of $\{\epsilon_{ij} \mid 1 \leq i < j \leq N\}$ and Hoeffding, we immediately conclude that there exist constant $E_8 > 0$ such that with probability at least $1 - n^{-1/2}$:

$$\sum_{j \in \mathcal{S}^c} \epsilon_{ij} \leq E_8 \sqrt{N \log N} \quad (\text{C.19})$$

and

$$\text{Var} \left(\sum_{j \in \mathcal{S}^c} \epsilon_{ij} \right) = O(N \log N) \quad (\text{C.20})$$

Combining (C.8), (C.9), (C.10), (C.16), (C.17), (C.18), and (C.20), we have

$$\text{Var} \left(\sum_{j \in \mathcal{S}^c} \hat{A}_{ij} - A_{ij} \right) = N^2 O \left(\frac{h^4}{n h^{d_\zeta}} + h^2 \delta_{N,n}^2 + \frac{1}{n} + \frac{1}{n^2 h^{2d_\zeta}} \right) \quad (\text{C.21})$$

In addition, combining the bias we derived previously, (B.8), (B.15), (C.5), (C.6), (C.7), (C.13), (C.14), (C.15), and (C.19), we can find constants $\bar{E}_1 > 0$ and $c > 0$ such that

$$\mathbb{P} \left(\left| \sum_{j \in \mathcal{S}^c} (\hat{A}_{ij} - A_{ij}) \right| \leq \bar{E}_1 N \left(h^2 + \frac{h^2}{\sqrt{n h^{d_\zeta}}} + h \delta_{N,n} + \frac{1}{\sqrt{n}} + \frac{1}{n h^{d_\zeta}} \right) \right) \geq 1 - c n^{-1/2}$$

When W is uniformly bounded, we can extend the analysis and obtain similar results. ■

D Supplementary Proofs

Lemma D.1 Let $\zeta_i \in \{\bar{\zeta}_1, \dots, \bar{\zeta}_G\}$ be i.i.d. random variables such that $0 \leq \bar{\zeta}_g \leq 1$ for any $g = 1, 2, \dots, G$, then $f(\cdot, \cdot)$ satisfies Assumption 2 except in the degenerate case where there exist $g \neq g'$ such that $f(\bar{\zeta}_g, \zeta) = f(\bar{\zeta}_{g'}, \zeta)$ for all $\zeta \in \{\bar{\zeta}_1, \dots, \bar{\zeta}_G\}$.

Proof of Lemma D.1. Without loss of generality, suppose that $f(\bar{\zeta}_1, \zeta_{g'}) > 0$ for all g' , then for any $g, g' \in \{1, \dots, G\}$,

$$\begin{aligned} & \max_{\zeta \in \{\bar{\zeta}_1, \dots, \bar{\zeta}_G\}} \left| \int f(\zeta, \zeta') (f(\bar{\zeta}_g, \zeta') - f(\bar{\zeta}_{g'}, \zeta')) d\mathbb{P}(\zeta') \right| \\ & \geq \max \left\{ \left| \int f(\bar{\zeta}_g, \zeta') (f(\bar{\zeta}_g, \zeta') - f(\bar{\zeta}_{g'}, \zeta')) d\mathbb{P}(\zeta') \right|, \left| \int f(\bar{\zeta}_{g'}, \zeta') (f(\bar{\zeta}_g, \zeta') - f(\bar{\zeta}_{g'}, \zeta')) d\mathbb{P}(\zeta') \right| \right\} \end{aligned}$$

Then expression above equals 0 if and only if

$$\int f^2(\bar{\zeta}_g, \zeta) d\mathbb{P}(\zeta) = \int f^2(\bar{\zeta}_{g'}, \zeta) d\mathbb{P}(\zeta) = \int f(\bar{\zeta}_g, \zeta) f(\bar{\zeta}_{g'}, \zeta) d\mathbb{P}(\zeta)$$

which implies that $f(\bar{\zeta}_g, \zeta) = f(\bar{\zeta}_{g'}, \zeta)$ for all $\zeta \in \{\bar{\zeta}_1, \dots, \bar{\zeta}_G\}$. ■

Lemma D.2 Let $\zeta_i \in [0, 1]^{d_\zeta}$ be i.i.d. random vectors such that the components of $\zeta_i = (\zeta_{i1}, \dots, \zeta_{id_\zeta})'$ are mutually independent and each follows the uniform distribution on $[0, 1]$, then

$$f(\zeta_i, \zeta_j) = \frac{\exp(\omega(\zeta_i, \zeta_j)' \beta)}{1 + \exp(\omega(\zeta_i, \zeta_j)' \beta)}$$

satisfies Assumption 2 when

$$\omega(\zeta_i, \zeta_j)' \beta = \zeta_{i1} + \zeta_{j1} + \sum_{k=2}^d \beta_k |\zeta_{ik} - \zeta_{jk}| + \sum_{k=d+1}^{d_\zeta} \beta_k (\zeta_{ik} - \zeta_{jk})^2, \quad \beta_2, \dots, \beta_k \leq 0$$

Proof of Lemma D.2. Since $\zeta_i \in [0, 1]^{d_\zeta}$ be i.i.d. random vectors such that the components of $\zeta_i = (\zeta_{i1}, \dots, \zeta_{id_\zeta})'$ are mutually independent and each follows the uniform distribution on $[0, 1]$, it is sufficiently to check the Assumption 2 under the following three specific case.

Case 1: $d_\zeta = 1$ and $\omega(\zeta_i, \zeta_j) = \zeta_i + \zeta_j$ In this case, for each $\zeta_1 < \zeta_2 \in [0, 1]$ we have

$$f(\zeta_1, \zeta_2) = \frac{\exp(\zeta_1 + \zeta_2)}{1 + \exp(\zeta_1 + \zeta_2)}$$

and

$$\begin{aligned} d_{12} &= \sup_{\zeta} \left| \int f(\zeta, \zeta') (f(\zeta_1, \zeta') - f(\zeta_2, \zeta')) d\zeta' \right| \\ &\geq \left| \int f(0, \zeta') (f(\zeta_1, \zeta') - f(\zeta_2, \zeta')) d\zeta' \right| \\ &\geq \left(\inf_{\zeta \in [0, 1]} f(0, \zeta) \right) \left| \int_0^1 \int_{\zeta_1}^{\zeta_2} f'(\zeta, \zeta') d\zeta d\zeta' \right| \\ &\stackrel{(i)}{\geq} \left(\inf_{\zeta \in [0, 1]} f(0, \zeta) \right) \left(\inf_{\zeta, \zeta' \in [0, 1]} f'(\zeta, \zeta') \right) |\zeta_2 - \zeta_1| \end{aligned}$$

where inequality (i) holds because $f'(\cdot, \cdot) > 0$ on $[0, 1]^2$. It is straightforward to check that $(\inf_{\zeta \in [0, 1]} f(0, \zeta)) > 0$ and $(\inf_{\zeta, \zeta' \in [0, 1]} f'(\zeta, \zeta')) > 0$, thus, let $\lambda = (\inf_{\zeta \in [0, 1]} f(0, \zeta)) (\inf_{\zeta, \zeta' \in [0, 1]} f'(\zeta, \zeta'))$, we prove that

$$d_{12} \geq \lambda |\zeta_2 - \zeta_1|$$

Case 2: $d_\zeta = 1$ and $\omega(\zeta_i, \zeta_j) = -|\zeta_i - \zeta_j|$ In this case, for each $\zeta_1 < \zeta_2 \in [0, 1]$ we have

$$f(\zeta_1, \zeta_2) = \frac{\exp(-|\zeta_1 - \zeta_2|)}{1 + \exp(-|\zeta_1 - \zeta_2|)}$$

and define

$$h_0(\zeta) = \int f(0, \zeta') f(\zeta, \zeta') d\zeta', \quad h_1(\zeta) = \int f(1, \zeta') f(\zeta, \zeta') d\zeta'$$

and it follows that

$$d_{12} = \sup_{\zeta} \left| \int f(\zeta, \zeta') (f(\zeta_1, \zeta') - f(\zeta_2, \zeta')) d\zeta' \right| \geq \max \{ |h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)| \}$$

Thus, it suffices to prove that there exists a constant $\lambda > 0$ such that for any $\zeta_1, \zeta_2 \in [0, 1]$:

$$\max \{ |h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)| \} \geq \lambda |\zeta_1 - \zeta_2|$$

It is straightforward to check that

$$\begin{aligned} h'_0(\zeta) &= - \int f(0, \zeta') \text{sign}(\zeta - \zeta') f(\zeta, \zeta') (1 - f(\zeta, \zeta')) d\zeta' \\ h_0(\zeta) &= h_1(1 - \zeta), \quad h'_1(\zeta) = -h'_0(1 - \zeta) \end{aligned}$$

Also, $h_0(\cdot)$ achieves its maximum at $\zeta_0^* < 1/2$ and $h_1(\cdot)$ achieves its maximum at $\zeta_1^* = 1 - \zeta_0^* > 1/2$. $h_0(\cdot)$ is a strictly increasing function ζ when $\zeta < \zeta_0^*$, and is a strictly decreasing function ζ when $\zeta > \zeta_0^*$.

Step 1 $\zeta_1 \in [\zeta_0^*, \zeta_1^*] \vee \zeta_2 \in [\zeta_0^*, \zeta_1^*]$ Fix an arbitrary positive constant $\delta < 1/2 - \zeta_0^*$, define

$$\lambda_0 = \inf_{\zeta \in [\zeta_0^* + \delta, 1]} |h'_0(\zeta)|$$

One can check that $\lambda_0 > 0$ because $h'_0(\cdot)$ is strictly negative on $[\zeta_0^* + \delta, 1]$. Then:

- (i) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_0^*, \zeta_1^* - \delta_0]$, we have $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0 |\zeta_1 - \zeta_2|$ by mean value theorem.
- (ii) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_1^* - \delta_0, \zeta_1^*]$, $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0 (\zeta_1^* - \zeta_2^* - \delta)$ and thus, $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0 (\zeta_1^* - \zeta_2^* - \delta) |\zeta_1 - \zeta_2|$ because $|\zeta_1 - \zeta_2| \leq 1$.
- (iii) When $\zeta_1, \zeta_2 \in [\zeta_0^*, \zeta_1^*]$, since $\delta < 1/2 - \zeta_0^*$, it follows that

$$\max \{ |h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)| \} \geq \lambda_0 |\zeta_1 - \zeta_2|$$

- (iv) When $\zeta_1 \in [\zeta_0^*, \zeta_0^* + \delta]$ and $\zeta_2 \in [\zeta_1^*, 1]$, $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0 (\zeta_1^* - \zeta_2^* - \delta)$ and thus, $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0 (\zeta_1^* - \zeta_2^* - \delta) |\zeta_1 - \zeta_2|$ because $|\zeta_1 - \zeta_2| \leq 1$.
- (v) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_0^*, \zeta_1^* - \delta_0]$, we have $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0 |\zeta_1 - \zeta_2|$ by mean value theorem.

Therefore, we can show that

$$\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda_0(\zeta_1^* - \zeta_0^* - \delta)|\zeta_1 - \zeta_2|$$

Step 2 $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$ Define

$$\lambda_1 = \inf_{\zeta \in [0, \zeta_0^*]} |h_0(\zeta) - h_1(\zeta)|$$

We have $\lambda_1 > 0$ because $h_0(\zeta) > h_1(\zeta)$ when $\zeta \in [0, \zeta_0^*]$. Thus, for any $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$, we have

(i) When $|h_0(\zeta_1) - h_0(\zeta_2)| \leq \lambda_1$, we have $|h_1(\zeta_1) - h_1(\zeta_2)| \geq 2\lambda_1 - \lambda_1 = \lambda_1$.

(ii) When $|h_1(\zeta_1) - h_1(\zeta_2)| \leq \lambda_1$, we have $|h_0(\zeta_1) - h_0(\zeta_2)| \geq 2\lambda_1 - \lambda_1 = \lambda_1$.

Therefore, we have $\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda_1 \geq \lambda_1|\zeta_1 - \zeta_2|$ when $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$.

Combining the conclusion in two steps yields that

$$d_{12} \geq \max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \min\{\lambda_0(\zeta_1^* - \zeta_0^* - \delta), \lambda_1\}|\zeta_2 - \zeta_1|$$

Case 3: $d_\zeta = 1$ and $\omega(\zeta_i, \zeta_j) = -(\zeta_i - \zeta_j)^2$ In this case, for each $\zeta_1 < \zeta_2 \in [0, 1]$ we have

$$f(\zeta_1, \zeta_2) = \frac{\exp(-(\zeta_1 - \zeta_2)^2)}{1 + \exp(-(\zeta_1 - \zeta_2)^2)}$$

and define

$$h_0(\zeta) = \int f(0, \zeta')f(\zeta, \zeta')d\zeta', \quad h_1(\zeta) = \int f(1, \zeta')f(\zeta, \zeta')d\zeta'$$

and it follows that

$$d_{12} = \sup_{\zeta} \left| \int f(\zeta, \zeta')(f(\zeta_1, \zeta') - f(\zeta_2, \zeta'))d\zeta' \right| \geq \max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\}$$

Thus, it suffices to prove that there exists a constant $\lambda > 0$ such that for any $\zeta_1, \zeta_2 \in [0, 1]$:

$$\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda|\zeta_1 - \zeta_2|$$

It is straightforward to check that

$$\begin{aligned} h'_0(\zeta) &= -2 \int f(0, \zeta')(\zeta - \zeta')f(\zeta, \zeta')(1 - f(\zeta, \zeta'))d\zeta' \\ h_0(\zeta) &= h_1(1 - \zeta), \quad h'_1(\zeta) = -h'_0(1 - \zeta) \end{aligned}$$

Also, $h_0(\cdot)$ achieves its maximum at $\zeta_0^* < 1/2$ and $h_1(\cdot)$ achieves its maximum at $\zeta_1^* = 1 - \zeta_0^* > 1/2$. $h_0(\cdot)$ is a strictly increasing function ζ when $\zeta < \zeta_0^*$, and is a strictly decreasing function ζ when $\zeta > \zeta_0^*$.

Step 1 $\zeta_1 \in [\zeta_0^*, \zeta_1^*] \vee \zeta_2 \in [\zeta_0^*, \zeta_1^*]$ Fix an arbitrary positive constant $\delta < 1/2 - \zeta_0^*$, define

$$\lambda_0 = \inf_{\zeta \in [\zeta_0^* + \delta, 1]} |h'_0(\zeta)|$$

One can check that $\lambda_0 > 0$ because $h'_0(\cdot)$ is strictly negative on $[\zeta_0^* + \delta, 1]$. Then:

- (i) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_0^*, \zeta_1^* - \delta_0]$, we have $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0 |\zeta_1 - \zeta_2|$ by mean value theorem.
- (ii) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_1^* - \delta_0, \zeta_1^*]$, $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0(\zeta_1^* - \zeta_2^* - \delta)$ and thus, $|h_1(\zeta_1) - h_1(\zeta_2)| \geq \lambda_0(\zeta_1^* - \zeta_2^* - \delta) |\zeta_1 - \zeta_2|$ because $|\zeta_1 - \zeta_2| \leq 1$.
- (iii) When $\zeta_1, \zeta_2 \in [\zeta_0^*, \zeta_1^*]$, since $\delta < 1/2 - \zeta_0^*$, it follows that

$$\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda_0 |\zeta_1 - \zeta_2|$$

- (iv) When $\zeta_1 \in [\zeta_0^*, \zeta_0^* + \delta]$ and $\zeta_2 \in [\zeta_1^*, 1]$, $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0(\zeta_1^* - \zeta_2^* - \delta)$ and thus, $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0(\zeta_1^* - \zeta_2^* - \delta) |\zeta_1 - \zeta_2|$ because $|\zeta_1 - \zeta_2| \leq 1$.
- (v) When $\zeta_1 \in [0, \zeta_0^*]$ and $\zeta_2 \in [\zeta_0^*, \zeta_1^* - \delta_0]$, we have $|h_0(\zeta_1) - h_0(\zeta_2)| \geq \lambda_0 |\zeta_1 - \zeta_2|$ by mean value theorem.

Therefore, we can show that

$$\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda_0(\zeta_1^* - \zeta_0^* - \delta) |\zeta_1 - \zeta_2|$$

Step 2 $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$ Define

$$\lambda_1 = \inf_{\zeta \in [0, \zeta_0^*]} |h_0(\zeta) - h_1(\zeta)|$$

We have $\lambda_1 > 0$ because $h_0(\zeta) > h_1(\zeta)$ when $\zeta \in [0, \zeta_0^*]$. Thus, for any $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$, we have

- (i) When $|h_0(\zeta_1) - h_0(\zeta_2)| \leq \lambda_1$, we have $|h_1(\zeta_1) - h_1(\zeta_2)| \geq 2\lambda_1 - \lambda_1 = \lambda_1$.
- (ii) When $|h_1(\zeta_1) - h_1(\zeta_2)| \leq \lambda_1$, we have $|h_0(\zeta_1) - h_0(\zeta_2)| \geq 2\lambda_1 - \lambda_1 = \lambda_1$.

Therefore, we have $\max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \lambda_1 \geq \lambda_1 |\zeta_1 - \zeta_2|$ when $\zeta_1 \in [0, \zeta_0^*] \wedge \zeta_2 \in [\zeta_1^*, 1]$.

Combining the conclusion in two steps yields that

$$d_{12} \geq \max\{|h_0(\zeta_1) - h_0(\zeta_2)|, |h_1(\zeta_1) - h_1(\zeta_2)|\} \geq \min\{\lambda_0(\zeta_1^* - \zeta_0^* - \delta), \lambda_1\} |\zeta_2 - \zeta_1|$$

■

Lemma D.3 *There are two sets of independent sub-Gaussian random variables $\{X_i\}_{i=1}^N$ and $\{Y_i\}_{i=1}^N$, such that there exist $\sigma_X, \sigma_Y > 0$:*

$$\sup_{1 \leq i \leq N} \mathbb{E}(e^{\lambda X_i}) \leq e^{\frac{\lambda^2 \sigma_X^2}{2}}, \quad \sup_{1 \leq i \leq N} \mathbb{E}(e^{\lambda Y_i}) \leq e^{\frac{\lambda^2 \sigma_Y^2}{2}}, \quad \forall \lambda \in \mathbb{R}$$

$\{X_i\}_{i=1}^N$ and $\{Y_i\}_{i=1}^N$ are independent of each other. Then for all $\epsilon > 0$, we have:

$$\mathbb{P}\left(\left|\frac{1}{N} \sum_{i=1}^N \alpha_i X_i\right| \geq \epsilon\right) \leq 2 \exp\left(-\frac{N \epsilon^2}{2 \sigma^2 \rho_\alpha^2 + \epsilon}\right)$$

In addition, suppose there is a set of independent random variables sub-Gaussian random variables $\{Y_i\}_{i=1}^N$ which is also independent of $\{X_i\}_{i=1}^N$ such that there exists $\sigma^2 > 0$:

$$\mathbb{P} \left(\left| \frac{1}{N} \sum_{i=1}^N X_i Y_i \right| \geq \epsilon \right) \leq 2 \exp \left(- \frac{N \epsilon^2}{2 (\sigma_X^2 \sigma_Y^2 + \sigma_X \sigma_Y \epsilon)} \right)$$

Proof. It is no harm to assume $\mathbb{E}(X_i) = \mathbb{E}(Y_i) = 0$ for all i . Then

$$\mathbb{E} \exp (\lambda X_i Y_i) = \mathbb{E} (\mathbb{E} (\exp (\lambda X_i Y_i) \mid X_i)) \leq \mathbb{E} \exp \left(\frac{\lambda^2 X_i^2 \sigma_Y^2}{2} \right) \leq \frac{1}{\sqrt{1 - \lambda^2 \sigma_Y^2 \sigma_X^2}}, \quad \forall 0 \leq \lambda^2 \leq \frac{1}{\sigma_Y^2 \sigma_X^2}$$

When $\lambda^2 < \frac{1}{\sigma_Y^2 \sigma_X^2}$, we have

$$\frac{1}{\sqrt{1 - \lambda^2 \sigma_X^2 \sigma_Y^2}} \leq \exp \left(\frac{\lambda^2 \sigma_X^2 \sigma_Y^2}{2(1 - \lambda^2 \sigma_X^2 \sigma_Y^2)} \right) \leq \exp \left(\frac{\lambda^2 \sigma_X^2 \sigma_Y^2}{2(1 - \lambda \sigma_X \sigma_Y)} \right)$$

Therefore, $X_i Y_i$ is satisfies Bernstein's condition because

$$\mathbb{E} (\lambda X_i Y_i) \leq \exp \left(\frac{\lambda^2 \sigma_X^2 \sigma_Y^2}{2(1 - \lambda \sigma_X \sigma_Y)} \right), \quad \forall |\lambda| < \frac{1}{\sigma_X \sigma_Y}$$

It immediately follows that

$$\mathbb{E} \left(\frac{1}{N} \sum_{i=1}^N \lambda X_i Y_i \right) \leq \exp \left(\frac{\lambda^2 \sigma_X^2 \sigma_Y^2}{2N(1 - \lambda \sigma_X \sigma_Y)} \right), \quad \forall |\lambda| < \frac{1}{\sigma_X \sigma_Y}$$

and

$$\mathbb{P} \left(\left| \frac{1}{N} \sum_{i=1}^N X_i Y_i \right| \geq \epsilon \right) \leq 2 \exp \left(- \frac{N \epsilon^2}{2 (\sigma_X^2 \sigma_Y^2 + \sigma_X \sigma_Y \epsilon)} \right)$$

■